

Notes on stability theory

Preliminary notes prepared by J. Iovino

DEPARTMENT OF MATHEMATICS
THE UNIVERSITY OF TEXAS AT SAN ANTONIO

Contents

Chapter 1. Stable formulas, both 2-valued and real-valued	5
1.1. The 2-valued case: φ -types and type definability	5
1.2. The real-valued case	8
1.3. Historical remarks	15
Chapter 2. Stability, trees, and definability of types	19
2.1. The equivalence proved in this chapter	19
2.2. From the absence of the order property to type definability	20
2.3. From the order property to tall trees	22
2.4. Tall trees produce non-definable types	23
2.5. Type spaces over countable models: a perfect set dichotomy	24
2.6. Historical remarks	26
Chapter 3. More stability, trees, and definability	27
3.1. A quantitative combinatorial proof (using Ramsey's theorem) that trees give the order property	27
3.2. The classical proof of definability, via ranks	29
3.3. Historical remarks	31
Chapter 4. Stability and notions of independence	33
4.1. Conventions	33
4.2. Definitions and heirs	34
4.3. Independence relations	37
4.4. Remarks	39
4.5. Historical remarks	40
Chapter 5. From stability to NIP	43
5.1. The independence property	43
5.2. The real-valued case	45
5.3. Historical remarks	47
Appendix A. Grothendieck's double limit criterion	49
A.1. The criterion	49
A.2. Historical remarks	52
Bibliography	53

Chapter 1

Stable formulas, both 2-valued and real-valued

Stability is a property of a single formula $\varphi(x, y)$, and this chapter looks at stability from two sides. On the logical side, a stable formula is one that cannot order arbitrarily long sequences of elements, and the central fact about such formulas is that all their types are definable: Section 1.1 sets up this side and states the definability theorem, Theorem 1.1.6, whose proof occupies Chapter 2. On the analytic side, a formula is simply a $\{0, 1\}$ -valued function of two variables, and nothing prevents us from replacing such functions by bounded real-valued ones. Once we do, stability becomes the statement that two iterated limits can be exchanged, and in Section 1.2 we show how closely this property is tied to reflexive Banach spaces.

1.1. The 2-valued case: φ -types and type definability

Throughout this section and Chapters 2 and 3, we fix a first-order theory T in a language L and an L -formula $\varphi(x, y)$ consistent with T , where x and y are finite tuples of variables. Tuples of elements are written $a, b, \dots$ and are always of the appropriate length. We write $\varphi^1 = \varphi$ and $\varphi^0 = \neg\varphi$.

Definition 1.1.1. Let M be a model of T .

- (a) A φ -formula over M is a formula $\varphi(x, b)^u$ with $b \in M$ and $u \in \{0, 1\}$.
- (b) A complete φ -type over M (in the variables x) is a set p of φ -formulas over M such that
 - (i) the set p is consistent with the elementary diagram of M , and
 - (ii) for each $b \in M$, exactly one of $\varphi(x, b)$ and $\neg\varphi(x, b)$ belongs to p .

The set of complete φ -types over M will be denoted $S_\varphi(M)$.

If $N \succeq M$ and $c \in N$, the set

$$\text{tp}_\varphi(c/M) = \{\varphi(x, b)^u : b \in M, u \in \{0, 1\}, N \models \varphi(c, b)^u\}$$

is a complete φ -type over M . Conversely, every complete φ -type p over M is consistent with the elementary diagram of M , so it is realized by some element c of some $N \succeq M$, and then $p = \text{tp}_\varphi(c/M)$.

For $p \in S_\varphi(M)$ we write $D_p = \{b \in M : \varphi(x, b) \in p\}$. By condition (ii), we have $p = \{\varphi(x, b) : b \in D_p\} \cup \{\neg\varphi(x, b) : b \in M \setminus D_p\}$, so p is determined by D_p . The set D_p records, for each parameter b , whether $\varphi(x, b)$ holds of the elements that realize p .

Lemma 1.1.2 (finite satisfiability). *Let $p \in S_\varphi(M)$ be given. For every finite set $\{\varphi(x, b_l)^{u_l} : l < k\} \subseteq p$ there is $a \in M$ with $M \models \bigwedge_{l < k} \varphi(a, b_l)^{u_l}$.*

Proof. The $L(M)$ -sentence $\exists x \bigwedge_{l < k} \varphi(x, b_l)^{u_l}$ is consistent with the elementary diagram of M . That diagram is a complete $L(M)$ -theory, so the sentence belongs to the diagram, that is, holds in M . ☺

Lemma 1.1.2 gives φ -types a natural topological home. The sets $\varphi(M, b) = \{a \in M^{|x|} : M \models \varphi(a, b)\}$, for $b \in M^{|y|}$, generate a Boolean algebra $\mathcal{B}_\varphi(M)$ of subsets of $M^{|x|}$. For $p \in S_\varphi(M)$, the sets $\varphi(M, b)$ with $\varphi(x, b) \in p$ and the complements $M^{|x|} \setminus \varphi(M, b)$ with $\neg\varphi(x, b) \in p$ have the finite intersection property by Lemma 1.1.2, and they generate an ultrafilter on $\mathcal{B}_\varphi(M)$. Conversely, an ultrafilter on $\mathcal{B}_\varphi(M)$ determines a set of φ -formulas over M that is finitely satisfiable in M , hence consistent with the elementary diagram of M , and so a complete φ -type. A complete φ -type over M is therefore the same thing as an ultrafilter on $\mathcal{B}_\varphi(M)$, and we regard $S_\varphi(M)$ as the Stone space of $\mathcal{B}_\varphi(M)$: a compact, Hausdorff, zero-dimensional space whose basic clopen sets consist of the types containing a given finite set of φ -formulas over M .

Definition 1.1.3. Let $p \in S_\varphi(M)$ be given.

- (a) The type p is *definable* if there is an $L(M)$ -formula $\psi(y)$, a φ -*definition* of p , such that for all $b \in M$, we have

$$\varphi(x, b) \in p \iff M \models \psi(b).$$

Equivalently, the set $D_p = \psi(M)$ is a definable subset of M .

- (b) A *positive Boolean combination of instances of φ over M* is a formula of the form

$$\bigvee_{i < n} \bigwedge_{a \in W_i} \varphi(a, y),$$

where $n < \omega$ and $W_0, \dots, W_{n-1}$ are finite subsets of M . By convention the empty disjunction is $\perp$ and the empty conjunction is $\top$.

By distributivity, every formula built from formulas $\varphi(a, y)$ ($a \in M$) using only $\wedge$ and $\vee$ is equivalent to a positive Boolean combination of instances of φ over M . If φ is quantifier-free, so is every such combination.

In general the set D_p is just a subset of M , with no reason to be describable inside M : when M is infinite there are far more subsets of M than formulas over M , and a definable type is one whose set D_p happens to be captured by a single formula. Realized types are the model case, since for $c \in M$ the type $\text{tp}_\varphi(c/M)$ is defined by the formula $\varphi(c, y)$. A definable type thus behaves like the type of an element of M , even when the type is realized only in an elementary extension.

Definability typically fails because of an order. In $(\mathbb{Q}, <)$, let $\varphi(x, y)$ be $x < y$ and let p be the type of $\sqrt{2}$ over $\mathbb{Q}$. The set $D_p = \{b \in \mathbb{Q} : b > \sqrt{2}\}$ is a cut with an irrational edge, whereas the subsets of $\mathbb{Q}$ definable in $(\mathbb{Q}, <)$ with parameters are finite unions of intervals with rational or infinite endpoints; so the type p is not definable. The type of an element just above a rational q , by contrast, is defined by $y > q$.

Definability also has a practical side. A formula $\psi(y)$ makes sense in every elementary extension $N \succeq M$, so a definition of p tells us how to extend p to parameters in N : put $\varphi(x, b)$ in the extension exactly when $N \models \psi(b)$. This is the source of the canonical extensions used to define independence in Chapter 4.

In $(\mathbb{Q}, <)$ the cut at $\sqrt{2}$ escapes every formula because the formula $x < y$ orders the rationals. The next definition isolates this behavior in finite pieces.

Definition 1.1.4. Let M be a model of T , and let $n \geq 1$ be given. An *n -ladder for φ in M* consists of tuples $a_1, \dots, a_n$ and $b_1, \dots, b_n$ of elements of M such that, for all $1 \leq i, j \leq n$, we have

$$M \models \varphi(a_i, b_j) \iff i \leq j.$$

We say that φ has the *order property* if for every n some model of T contains an n -ladder for φ , and that φ is *stable* otherwise.

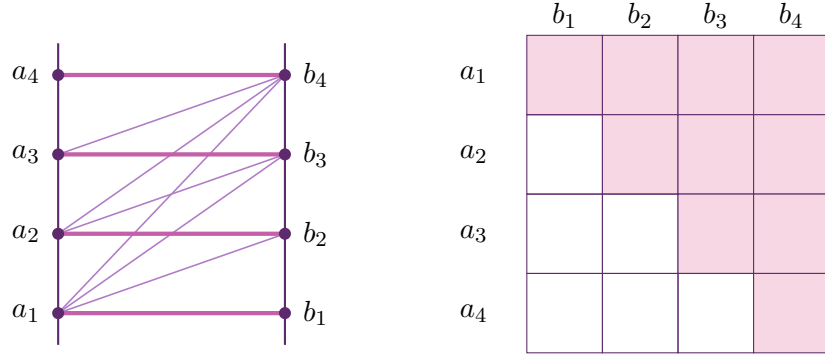


FIGURE 1.1. A 4-ladder. Left: the elements a_i and b_j are joined when $\varphi(a_i, b_j)$ holds, i.e. when $i \leq j$. The pairs (a_i, b_i) are the rungs (thick, pink), and each a_i also reaches every b_j with $j > i$ (purple). Right: the truth table of $\varphi(a_i, b_j)$, with pink cells where φ holds. The table is a staircase, which is why an n -ladder is also called a *half-graph*.

Figure 1.1 explains the name. Put the a_i on one rail and the b_j on the other, both in increasing order, and join a_i to b_j whenever $\varphi(a_i, b_j)$ holds; the “horizontal” pairs (a_i, b_i) are then the rungs of a ladder. What matters is the asymmetry between the two directions: each a_i is related to every b_j with $j \geq i$ and to none with $j < i$, so the formula φ linearly orders the rungs, in the sense that $i \leq j$ if and only if $\varphi(a_i, b_j)$.



Remark 1.1.5. An $(n+1)$ -ladder contains an n -ladder. Hence, by compactness, the formula φ has the order property if and only if some model of T contains an infinite sequence $(a_i, b_i)_{i < \omega}$ with $\varphi(a_i, b_j) \iff i \leq j$. The choice of “ $\leq$ ” is immaterial. Suppose that $a_1, \dots, a_n$ and $b_1, \dots, b_n$ are tuples in a model of T such that $\varphi(a_i, b_j) \iff i > j$ for $1 \leq i, j \leq n$. Reversing the indices by putting $a'_i = a_{n+1-i}$ and $b'_j = b_{n+1-j}$ gives $\varphi(a'_i, b'_j) \iff i < j$. Shifting by putting $b''_j = b'_{j+1}$ then gives $\varphi(a'_i, b''_j) \iff i \leq j$ for $1 \leq i, j \leq n-1$. So a “ $>$ ”-pattern of length n yields an $(n-1)$ -ladder.

The main theorem of this chapter confirms that the order property is the only obstruction to definability, and that when the obstruction is absent the definitions can be chosen of the simplest possible kind: positive Boolean combinations of instances of φ (see Definition 1.1.3(b)).

Theorem 1.1.6 (Definability of types). *The following are equivalent.*

- (1) *The formula φ does not have the order property.*
- (2) *For every model M of T , every $p \in S_\varphi(M)$ has a φ -definition that is a positive Boolean combination of instances $\varphi(a, y)$ with $a \in M$.*
- (3) *For every model M of T , every $p \in S_\varphi(M)$ is definable.*

In short: *all φ -types are definable if and only if φ does not have the order property.* The implication (2) $\Rightarrow$ (3) is trivial, and the implication (1) $\Rightarrow$ (2) is proved in Section 2.2. The remaining implication (3) $\Rightarrow$ (1) takes a detour through binary trees: in Sections 2.3–2.4 we show that if φ has the order property, then the formula φ encodes tall trees, and that tall trees produce a φ -type that is not definable.

Chapter Roadmap

The rest of this chapter, Section 1.2, looks at stability from the analytic side: there a formula becomes a bounded real-valued function, stability becomes the exchange of two iterated limits, and spaces of types become compact spaces of the kind studied in functional analysis.

Chapters 2 and 3 do not depend on Section 1.2. A reader mainly interested in the classical theory may go directly to Chapter 2, where Theorem 1.1.6 is proved, and return to Section 1.2 later; the analytic proof of (1) $\Rightarrow$ (3) given there (see Corollary 1.2.13) is perhaps best appreciated after the combinatorial one. The two sides meet again in Chapter 4, where Harrington's lemma, the fact behind the symmetry of independence, turns out to be the exchange of iterated ultrafilter limits (see Subsection 4.2.1), and in Section 5.2, where the independence property is characterized in terms of convergent subsequences by means of Rosenthal's dichotomy.

1.2. The real-valued case

Once a formula is identified with a $\{0,1\}$ -valued function, stability can be read as a property of bounded real-valued functions, namely the exchangeability of iterated limits. We follow Iovino [49]. Seen this way, stability meets the reflexivity of Banach spaces, through a characterization of reflexivity due to James [50] that looks very much like the order property.

1.2.1. Stability and iterated limit exchange

Let M be an L -structure and $\varphi(x, y)$ an L -formula. We identify φ with the function $\varphi: M^{|x|} \times M^{|y|} \rightarrow \{0,1\}$ that takes the value 1 at (a, b) if and only if $M \models \varphi(a, b)$. More generally, a function $\varphi: A \times B \rightarrow \{0,1\}$ on arbitrary sets A, B has the *order property* if there are sequences $(a_m)_{m < \omega}$ in A and $(b_n)_{n < \omega}$ in B such that

$$\varphi(a_m, b_n) = 1 \iff m \leq n.$$

Definition 1.2.1. A function $\varphi: A \times B \rightarrow \{0,1\}$ is *unstable* if either φ or $\neg\varphi = 1 - \varphi$ has the order property, and *stable* otherwise. A formula $\varphi(x, y)$ is *stable on M* if the corresponding function on $M^{|x|} \times M^{|y|}$ is stable.



Remark 1.2.2. A formula $\varphi(x, y)$ is stable in the sense of Definition 1.1.4 if and only if φ is stable on every model of T . Indeed, if φ has the order property, then by compactness (see Remark 1.1.5) some model contains an infinite ladder, so φ is unstable on that model. Conversely, if φ or $\neg\varphi$ has the order property on some model M of T , then M contains n -ladders for φ for every n : directly in the first case, and by Remark 1.1.5 in the second, since $\neg\varphi(a_m, b_n) \iff m \leq n$ means $\varphi(a_m, b_n) \iff m > n$.

To state the analytic form of stability, we use limits along ultrafilters. For a family $(a_i)_{i \in I}$ in a topological space X , a point $a \in X$, and an ultrafilter $\mathcal{U}$ on I , we write

$$(1.1) \quad \lim_{i, \mathcal{U}} a_i = a$$

if a is a limit of (a_i) along $\mathcal{U}$. We use freely the standard facts about such limits: limits along ultrafilters are unique in Hausdorff spaces, always exist in compact spaces, describe closures, and are preserved by continuous maps.

Along an infinite ladder, the two iterated limits of φ along a nonprincipal ultrafilter on ω already differ: for fixed m the values $\varphi(a_m, b_n)$ are eventually 1, and for fixed n the values $\varphi(a_m, b_n)$ are eventually 0, so one iterated limit is 1 and the other is 0. This is the phenomenon

that the analytic form of stability forbids. Ultrafilters on arbitrary index sets are convenient, but ladders are indexed by ω . The next proposition, which we use several times, bridges the gap: from iterated limits taken along arbitrary families it extracts sequences on which the two iterated limits are already visible, one above the diagonal and the other below.

Proposition 1.2.3. *Let $\varphi: A \times B \rightarrow [0, 1]$ be a function, let $(a_i)_{i \in I}$ be a family in A and $(b_j)_{j \in J}$ a family in B , and let $\mathcal{U}, \mathcal{V}$ be ultrafilters on I, J with*

$$\lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}} \varphi(a_i, b_j) = \alpha, \quad \lim_{j, \mathcal{V}} \lim_{i, \mathcal{U}} \varphi(a_i, b_j) = \beta.$$

Then there are $i_0, i_1, \dots \in I$ and $j_0, j_1, \dots \in J$ such that

$$|\varphi(a_{i_m}, b_{j_n}) - \alpha| \leq 2^{1-m} \text{ if } m < n, \quad |\varphi(a_{i_m}, b_{j_n}) - \beta| \leq 2^{1-n} \text{ if } n < m.$$

Proof. By definition of the limits, for every $\varepsilon > 0$ there are $U_\varepsilon \in \mathcal{U}$ and $V_\varepsilon \in \mathcal{V}$ such that

$$i \in U_\varepsilon \implies \left| \lim_{j, \mathcal{V}} \varphi(a_i, b_j) - \alpha \right| \leq \varepsilon,$$

$$j \in V_\varepsilon \implies \left| \lim_{i, \mathcal{U}} \varphi(a_i, b_j) - \beta \right| \leq \varepsilon.$$

Also, for all $\hat{i} \in I, \hat{j} \in J$ and $\varepsilon > 0$ there are $V_\varepsilon^{\hat{i}} \in \mathcal{V}$ and $U_\varepsilon^{\hat{j}} \in \mathcal{U}$ such that

$$j \in V_\varepsilon^{\hat{i}} \implies \left| \varphi(a_{\hat{i}}, b_j) - \lim_{j, \mathcal{V}} \varphi(a_{\hat{i}}, b_j) \right| \leq \varepsilon,$$

$$i \in U_\varepsilon^{\hat{j}} \implies \left| \varphi(a_i, b_{\hat{j}}) - \lim_{i, \mathcal{U}} \varphi(a_i, b_{\hat{j}}) \right| \leq \varepsilon.$$

Choose recursively

$$i_N \in U_{2^{-N}} \cap \bigcap_{k < N} U_{2^{-k}}^{j_k}, \quad j_N \in V_{2^{-N}} \cap \bigcap_{k \leq N} V_{2^{-k}}^{i_k}.$$

These sets are finite intersections of members of an ultrafilter, hence nonempty. If $m < n$, then $j_n \in V_{2^{-m}}^{i_m}$ and $i_m \in U_{2^{-m}}$, so $|\varphi(a_{i_m}, b_{j_n}) - \alpha| \leq 2^{-n} + 2^{-m} \leq 2^{1-m}$. If $n < m$, then $i_m \in U_{2^{-n}}^{j_n}$ and $j_n \in V_{2^{-n}}$, so $|\varphi(a_{i_m}, b_{j_n}) - \beta| \leq 2^{-m} + 2^{-n} \leq 2^{1-n}$. ☺

Figure 1.2 shows the array produced by Proposition 1.2.3.

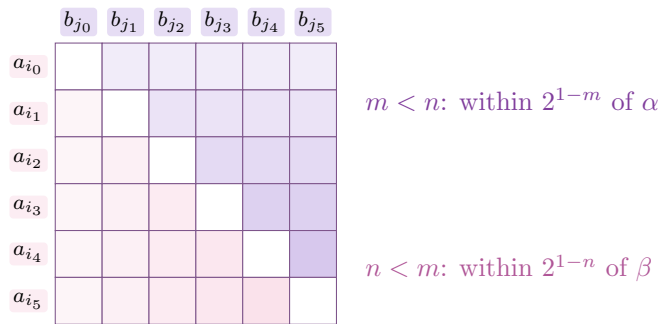


FIGURE 1.2. The array $\varphi(a_{i_m}, b_{j_n})$ of Proposition 1.2.3. Above the diagonal the values approach α , below it they approach β , and the deepening shades show the tolerances 2^{1-m} and 2^{1-n} shrinking. Nothing is claimed on the diagonal. When $\alpha \neq \beta$ the array is a ladder in the limit: compare the staircase of Figure 1.1.

Proposition 1.2.4. *Let $\varphi: A \times B \rightarrow \{0, 1\}$ be a function. The following are equivalent.*

- (1) *The function φ is stable.*
 (2) *For all sequences (a_m) in A and (b_n) in B and all ultrafilters $\mathcal{U}, \mathcal{V}$ on $\mathbb{N}$, we have*

$$\lim_{m, \mathcal{U}} \lim_{n, \mathcal{V}} \varphi(a_m, b_n) = \lim_{n, \mathcal{V}} \lim_{m, \mathcal{U}} \varphi(a_m, b_n).$$

Proof. (2) $\Rightarrow$ (1). Suppose that φ is unstable. If φ has the order property, there are sequences (a_m) in A and (b_n) in B with $\varphi(a_m, b_n) = 1 \iff m \leq n$. Let $\mathcal{U} = \mathcal{V}$ be a nonprincipal ultrafilter on $\mathbb{N}$. For fixed m we have $\varphi(a_m, b_n) = 1$ for all $n \geq m$, so the left-hand side of (2) is 1. For fixed n we have $\varphi(a_m, b_n) = 0$ for all $m > n$, so the right-hand side is 0. If instead $\neg\varphi$ has the order property, the two sides are 0 and 1.

(1) $\Rightarrow$ (2). Suppose that (2) fails: for some sequences $(a_m), (b_n)$ and ultrafilters $\mathcal{U}, \mathcal{V}$, the two iterated limits are $\alpha \neq \beta$. Since φ is $\{0, 1\}$ -valued, both limits lie in $\{0, 1\}$. By Proposition 1.2.3 there are sequences (a'_m) in A and (b'_n) in B with $|\varphi(a'_m, b'_n) - \alpha| \leq 2^{1-m}$ for $m < n$ and $|\varphi(a'_m, b'_n) - \beta| \leq 2^{1-n}$ for $n < m$. Since φ is $\{0, 1\}$ -valued, we have $\varphi(a'_m, b'_n) = \alpha$ for $2 \leq m < n$ and $\varphi(a'_m, b'_n) = \beta$ for $2 \leq n < m$. Put $c_m = a'_{2m+2}$ and $d_n = b'_{2n+3}$. Since $2m+2 < 2n+3 \iff m \leq n$, we get $\varphi(c_m, d_n) = \alpha$ if $m \leq n$ and $\varphi(c_m, d_n) = \beta$ if $m > n$. So φ has the order property if $\alpha = 1$, and $\neg\varphi$ does if $\alpha = 0$. $\odot$

Condition (2) of Proposition 1.2.4 makes sense for every bounded real-valued function, and we take this condition as the definition of stability in the real-valued setting.

Definition 1.2.5. A bounded function $\varphi: A \times B \rightarrow \mathbb{R}$ is *stable* if, for all sequences (a_m) in A and (b_n) in B and all ultrafilters $\mathcal{U}, \mathcal{V}$ on $\mathbb{N}$, we have

$$\lim_{m, \mathcal{U}} \lim_{n, \mathcal{V}} \varphi(a_m, b_n) = \lim_{n, \mathcal{V}} \lim_{m, \mathcal{U}} \varphi(a_m, b_n).$$

By Proposition 1.2.4, the new definition agrees with Definition 1.2.1 for $\{0, 1\}$ -valued functions, and by Proposition 1.2.3, the sequences in Definition 1.2.5 may be replaced by arbitrary families, as the next corollary records. For a bounded set of continuous functions on a compact space, the same condition is the hypothesis of Grothendieck's double limit criterion (see Appendix A).

Corollary 1.2.6. *If $\varphi: A \times B \rightarrow [0, 1]$ is stable, then $\lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}} \varphi(a_i, b_j) = \lim_{j, \mathcal{V}} \lim_{i, \mathcal{U}} \varphi(a_i, b_j)$ for all families $(a_i)_{i \in I}, (b_j)_{j \in J}$ and all ultrafilters $\mathcal{U}$ on I , $\mathcal{V}$ on J .*

Proof. If the two limits are $\alpha \neq \beta$, take $(i_m), (j_n)$ as in Proposition 1.2.3 and a nonprincipal ultrafilter $\mathcal{W}$ on $\mathbb{N}$. For fixed m , the limit $\lim_{n, \mathcal{W}} \varphi(a_{i_m}, b_{j_n})$ is within 2^{1-m} of α , so $\lim_{m, \mathcal{W}} \lim_{n, \mathcal{W}} \varphi(a_{i_m}, b_{j_n}) = \alpha$. Symmetrically, the other iterated limit is β , which contradicts the stability of φ . $\odot$

The proof of Corollary 1.2.6 used a single nonprincipal ultrafilter $\mathcal{W}$ on $\mathbb{N}$, in both coordinates, and this is no accident. For ultrafilters $\mathcal{U}$ on I and $\mathcal{V}$ on J , let $\mathcal{U} \otimes \mathcal{V}$ be the ultrafilter on $I \times J$ consisting of the sets X with $\{i : \{j : (i, j) \in X\} \in \mathcal{V}\} \in \mathcal{U}$. For every bounded $f: I \times J \rightarrow \mathbb{R}$ we have $\lim_{(i, j), \mathcal{U} \otimes \mathcal{V}} f(i, j) = \lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}} f(i, j)$, so the two iterated limits of Definition 1.2.5 are the limits of $\varphi(a_m, b_n)$ along $\mathcal{U} \otimes \mathcal{V}$ and along the transpose of $\mathcal{V} \otimes \mathcal{U}$. These two ultrafilters can differ, because the tensor product of ultrafilters is not commutative, and the simplest witness already lives on $\mathbb{N} \times \mathbb{N}$: if $\mathcal{W}$ is a nonprincipal ultrafilter on $\mathbb{N}$, then the upper triangle $\{(m, n) : m < n\}$ belongs to $\mathcal{W} \otimes \mathcal{W}$, because each of its rows is cofinite, while the lower triangle $\{(m, n) : m > n\}$ does not, because each of its rows is finite. A ladder is exactly a pair of sequences along which φ is the indicator function of the closed upper triangle $\{(m, n) : m \leq n\}$. So the order property says that φ can tell $\mathcal{W} \otimes \mathcal{W}$ from its own transpose, and the proofs of Proposition 1.2.4 and Corollary 1.2.6 show the following:

a bounded function φ is stable if and only if, for one (equivalently, for every) nonprincipal ultrafilter $\mathcal{W}$ on $\mathbb{N}$ and all sequences (a_m) and (b_n) , we have

$$\lim_{m, \mathcal{W}} \lim_{n, \mathcal{W}} \varphi(a_m, b_n) = \lim_{n, \mathcal{W}} \lim_{m, \mathcal{W}} \varphi(a_m, b_n).$$

Stable functions are blind to the noncommutativity of the tensor square of a single ultrafilter.

1.2.2. Stability and continuous extendability

Exchanging iterated limits is a statement about continuity in disguise. In Proposition 1.2.8 below we show that a function $\varphi: A \times B \rightarrow [0, 1]$ is stable exactly when the function extends to a separately continuous function on pairs of types. The types in question live in compact spaces built from A and B .

Definition 1.2.7. Let $\varphi: A \times B \rightarrow [0, 1]$ be a function. For $a \in A$, the *left φ -type* of a is the function $\text{ltp}_\varphi(a): y \mapsto \varphi(a, y)$ on B . For $b \in B$, the *right φ -type* of b is the function $\text{rtp}_\varphi(b): x \mapsto \varphi(x, b)$ on A . The *space of left φ -types* $\text{LS}(\varphi)$ is the closure of $\{\text{ltp}_\varphi(a) : a \in A\}$ in $[0, 1]^B$ with the product topology, and the *space of right φ -types* $\text{RS}(\varphi)$ is the closure of $\{\text{rtp}_\varphi(b) : b \in B\}$ in $[0, 1]^A$.

By Tychonoff's theorem, the spaces $[0, 1]^A$ and $[0, 1]^B$ are compact, and so are their closed subspaces $\text{RS}(\varphi)$ and $\text{LS}(\varphi)$. Both spaces are quotients of Stone–Čech compactifications. The map $a \mapsto \text{ltp}_\varphi(a)$ from the discrete space A into the compact space $[0, 1]^B$ extends to a continuous map $\beta A \rightarrow [0, 1]^B$, namely $\mathcal{U} \mapsto \lim_{a, \mathcal{U}} \text{ltp}_\varphi(a)$, whose image is compact and contains the realized types, and is therefore $\text{LS}(\varphi)$. So every left φ -type is the limit of the realized types along an ultrafilter on A , the principal ultrafilters give the realized types, and $\text{LS}(\varphi)$ is a continuous image of βA ; likewise $\text{RS}(\varphi)$ is a continuous image of βB .

Proposition 1.2.8. For $\varphi: A \times B \rightarrow [0, 1]$ the following are equivalent.

- (1) The function φ is stable.
- (2) There is a separately continuous function $\hat{\varphi}: \text{LS}(\varphi) \times \text{RS}(\varphi) \rightarrow [0, 1]$ such that

$$\hat{\varphi}(\text{ltp}_\varphi(a), \text{rtp}_\varphi(b)) = \varphi(a, b) \quad \text{for all } (a, b) \in A \times B.$$

Proof. (1) $\Rightarrow$ (2). Let $p \in \text{LS}(\varphi)$ and $q \in \text{RS}(\varphi)$ be given. There are families $(a_i)_{i \in I}$ in A , $(b_j)_{j \in J}$ in B and ultrafilters $\mathcal{U}$ on I , $\mathcal{V}$ on J with

$$p = \lim_{i, \mathcal{U}} \text{ltp}_\varphi(a_i), \quad q = \lim_{j, \mathcal{V}} \text{rtp}_\varphi(b_j).$$

By Corollary 1.2.6, we have $\lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}} \varphi(a_i, b_j) = \lim_{j, \mathcal{V}} \lim_{i, \mathcal{U}} \varphi(a_i, b_j)$. Define $\hat{\varphi}(p, q)$ to be this common value. Since limits in the product topology are computed coordinatewise, this common value satisfies

$$(1.2) \quad \hat{\varphi}(p, q) = \lim_{i, \mathcal{U}} q(a_i) = \lim_{j, \mathcal{V}} p(b_j).$$

$\hat{\varphi}$ is well defined. Suppose also $p = \lim_{i, \mathcal{U}'} \text{ltp}_\varphi(a'_i)$ and $q = \lim_{j, \mathcal{V}'} \text{rtp}_\varphi(b'_j)$. Then $\lim_{j, \mathcal{V}} \varphi(x, b_j) = q(x) = \lim_{j, \mathcal{V}'} \varphi(x, b'_j)$ for every $x \in A$, and $\lim_{i, \mathcal{U}} \varphi(a_i, y) = p(y) = \lim_{i, \mathcal{U}'} \varphi(a'_i, y)$ for every $y \in B$. Hence, using Corollary 1.2.6 for the middle step, we have

$$\lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}} \varphi(a_i, b_j) = \lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}'} \varphi(a_i, b'_j) = \lim_{j, \mathcal{V}'} \lim_{i, \mathcal{U}} \varphi(a_i, b'_j) = \lim_{j, \mathcal{V}'} \lim_{i, \mathcal{U}'} \varphi(a'_i, b'_j).$$

$\hat{\varphi}$ extends φ . Take constant families.

$\hat{\varphi}$ is separately continuous. Fix p , and let $q = \lim_{k, \mathcal{W}} q_k$ in $\text{RS}(\varphi)$, where $\mathcal{W}$ is an ultrafilter on a set K . Write $q_k = \lim_{j, \mathcal{V}_k} \text{rtp}_\varphi(b_{k,j})$ with j ranging over a set J_k . On $J' = \{(k, j) : k \in K, j \in J_k\}$ let $\mathcal{V}'$ be the ultrafilter of all $X \subseteq J'$ with $\{k : \{j : (k, j) \in X\} \in \mathcal{V}_k\} \in \mathcal{W}$. For

every family $(z_{k,j})$ in a compact Hausdorff space we have $\lim_{(k,j), \mathcal{V}'} z_{k,j} = \lim_{k, \mathcal{W}} \lim_{j, \mathcal{V}_k} z_{k,j}$. Hence $q = \lim_{(k,j), \mathcal{V}'} \text{rtp}_\varphi(b_{k,j})$, and by (1.2), we have

$$\hat{\varphi}(p, q) = \lim_{(k,j), \mathcal{V}'} p(b_{k,j}) = \lim_{k, \mathcal{W}} \lim_{j, \mathcal{V}_k} p(b_{k,j}) = \lim_{k, \mathcal{W}} \hat{\varphi}(p, q_k).$$

So $\hat{\varphi}(p, \cdot)$ is continuous. Continuity of $\hat{\varphi}(\cdot, q)$ is proved symmetrically.

(2) $\Rightarrow$ (1). Let (a_m) , (b_n) be sequences in A , B , let $\mathcal{U}, \mathcal{V}$ be ultrafilters on $\mathbb{N}$, and put $p = \lim_{m, \mathcal{U}} \text{ltp}_\varphi(a_m)$ and $q = \lim_{n, \mathcal{V}} \text{rtp}_\varphi(b_n)$. Then $(p, q) \in \text{LS}(\varphi) \times \text{RS}(\varphi)$, and by separate continuity

$$\lim_{m, \mathcal{U}} \lim_{n, \mathcal{V}} \varphi(a_m, b_n) = \lim_{m, \mathcal{U}} \hat{\varphi}(\text{ltp}_\varphi(a_m), q) = \hat{\varphi}(p, q) = \lim_{n, \mathcal{V}} \lim_{m, \mathcal{U}} \varphi(a_m, b_n). \quad \text{☺}$$

1.2.3. Stability and reflexivity

The pairing between a reflexive Banach space and its dual, restricted to the product of the two unit balls with their weak topologies, is a separately continuous function on a product of compact spaces. In Theorem 1.2.9 below we state Iovino's theorem that this example is universal: the stable functions are exactly the restrictions of the pairing between the unit balls of a reflexive Banach space E and of E^* .

For a Banach space E with dual E^* , we write B_E for the closed unit ball of E and $\langle x, y \rangle$ for $y(x)$, where $x \in E$ and $y \in E^*$. Recall that E is *reflexive* if the canonical embedding of E into E^{**} is surjective.

Theorem 1.2.9 (Iovino). *For $\varphi: A \times B \rightarrow [0, 1]$ the following are equivalent.*

- (1) *The function φ is stable.*
- (2) *There are a reflexive Banach space E and maps $u: A \rightarrow B_E$ and $v: B \rightarrow B_{E^*}$ such that*

$$\varphi(x, y) = \langle u(x), v(y) \rangle \quad \text{for all } (x, y) \in A \times B.$$

Proof. See [49, Theorem 3.3]. For (1) $\Rightarrow$ (2), stability is used to show that the operator $T: \ell^\infty(A)^* \rightarrow \ell^\infty(B)$ given by $T\mu(y) = \int \varphi(x, y) d\mu(x)$ is weakly compact; this step goes back to Krivine and Maurey [51] and Raynaud [52]. The theorem of Davis, Figiel, Johnson and Pełczyński [46] then factors T through a reflexive space. The implication (2) $\Rightarrow$ (1) follows from the weak compactness of B_E and the weak* compactness of B_{E^*} . ☺

Read back into model theory, the theorem says the following. By Remark 1.2.2 and Proposition 1.2.4, a formula $\varphi(x, y)$ is stable in the sense of Definition 1.1.4 if and only if, in every model M of T , the function φ on M is the restriction of the pairing map $B_E \times B_{E^*} \rightarrow \mathbb{R}$ for some reflexive Banach space E (depending on M), composed with maps $M^{|x|} \rightarrow B_E$ and $M^{|y|} \rightarrow B_{E^*}$.

1.2.4. Stability and polynomial approximation

By Theorem 1.1.6, if φ does not have the order property, every φ -type over a model of T is defined by a Boolean combination of instances $\varphi(a, y)$. On $\{0, 1\}$ the Boolean connectives are polynomials, namely $\neg t = 1 - t$ and $t \wedge s = ts$. The real-valued analogue of a definition is therefore a *uniform limit of polynomials in instances of φ* , and in Theorem 1.2.11 below we use the Stone–Weierstrass theorem to show that stability is exactly what is needed for such approximations to exist. Continuous logic treats definability in the same way, and Ben Yaacov used polynomial approximation to derive definability of types from Grothendieck's double limit criterion (see the historical remarks at the end of this chapter, and Appendix A). We use the Stone–Weierstrass theorem in the following form.

Fact 1.2.10 (Stone–Weierstrass). *Let K be a compact Hausdorff space and let $\mathcal{F} \subseteq C(K)$ be a family of continuous real-valued functions that separates the points of K . Then every $f \in C(K)$ is a uniform limit on K of functions of the form $P(g_1, \dots, g_k)$, where $g_1, \dots, g_k \in \mathcal{F}$ and P is a real polynomial in k variables.*

To apply the theorem we need continuous functions on the type spaces that separate points, and the instances of φ supply them. Fix $\varphi: A \times B \rightarrow [0, 1]$. For $a \in A$, let $e_a: \text{RS}(\varphi) \rightarrow [0, 1]$ be the evaluation map $e_a(q) = q(a)$, and for $b \in B$, let $e_b: \text{LS}(\varphi) \rightarrow [0, 1]$ be $e_b(p) = p(b)$. These maps are continuous, because $\text{RS}(\varphi)$ and $\text{LS}(\varphi)$ carry the product topology, and on realized types they are instances of φ , since $e_a(\text{rtp}_\varphi(b)) = \varphi(a, b) = e_b(\text{ltp}_\varphi(a))$. The family $\{e_a : a \in A\}$ separates the points of $\text{RS}(\varphi)$, since two distinct functions $q \neq q'$ in $[0, 1]^A$ differ at some $a \in A$, and likewise the family $\{e_b : b \in B\}$ separates the points of $\text{LS}(\varphi)$.

Theorem 1.2.11. *For $\varphi: A \times B \rightarrow [0, 1]$ the following are equivalent.*

- (1) *The function φ is stable.*
- (2) *There is a function $\hat{\varphi}: \text{LS}(\varphi) \times \text{RS}(\varphi) \rightarrow [0, 1]$ with $\hat{\varphi}(\text{ltp}_\varphi(a), \text{rtp}_\varphi(b)) = \varphi(a, b)$ for all $(a, b) \in A \times B$, such that for every $\varepsilon > 0$:*

- (a) *for every $p \in \text{LS}(\varphi)$ there are $a_1, \dots, a_k \in A$ and a real polynomial P with*

$$|\hat{\varphi}(p, q) - P(q(a_1), \dots, q(a_k))| \leq \varepsilon \quad \text{for all } q \in \text{RS}(\varphi);$$

- (b) *for every $q \in \text{RS}(\varphi)$ there are $b_1, \dots, b_l \in B$ and a real polynomial Q with*

$$|\hat{\varphi}(p, q) - Q(p(b_1), \dots, p(b_l))| \leq \varepsilon \quad \text{for all } p \in \text{LS}(\varphi).$$

Proof. (1) $\Rightarrow$ (2). Let $\hat{\varphi}$ be the separately continuous function given by Proposition 1.2.8. For fixed p , the function $\hat{\varphi}(p, \cdot)$ is continuous on the compact Hausdorff space $\text{RS}(\varphi)$. Applying Fact 1.2.10 with $\mathcal{F} = \{e_a : a \in A\}$ gives (a). Clause (b) follows in the same way, using the continuity of $\hat{\varphi}(\cdot, q)$ on $\text{LS}(\varphi)$ and $\mathcal{F} = \{e_b : b \in B\}$.

(2) $\Rightarrow$ (1). Each function $q \mapsto P(q(a_1), \dots, q(a_k))$ is continuous on $\text{RS}(\varphi)$. By (a), the function $\hat{\varphi}(p, \cdot)$ is a uniform limit of such functions, hence continuous. By (b), the function $\hat{\varphi}(\cdot, q)$ is continuous for the same reason. Hence $\hat{\varphi}$ is separately continuous, and φ is stable by Proposition 1.2.8. ☺

Evaluated at realized right types, clause (a) says that left types are approximated by polynomials in instances. Suppose that φ is stable, and let $\hat{\varphi}$ be the separately continuous extension given by Proposition 1.2.8, as in the proof of (1) $\Rightarrow$ (2). Then we have $\hat{\varphi}(p, \text{rtp}_\varphi(b)) = p(b)$ for $p \in \text{LS}(\varphi)$ and $b \in B$: if $p = \lim_{i, \mathcal{U}} \text{ltp}_\varphi(a_i)$, continuity in the first variable gives $\hat{\varphi}(p, \text{rtp}_\varphi(b)) = \lim_{i, \mathcal{U}} \varphi(a_i, b) = p(b)$. Taking $q = \text{rtp}_\varphi(b)$ in (a) therefore gives

$$|p(b) - P(\varphi(a_1, b), \dots, \varphi(a_k, b))| \leq \varepsilon \quad \text{for all } b \in B.$$

So every left φ -type is a uniform limit on B of polynomials in instances $\varphi(a, y)$ with $a \in A$, and symmetrically for right types.

The approximation in Theorem 1.2.11 is required on all of $\text{RS}(\varphi)$ (respectively $\text{LS}(\varphi)$), not only at realized types, and this requirement cannot be weakened: the next example shows that approximation on B alone does not imply stability, since there every type is trivially approximable on B and yet the function has the order property.

Example 1.2.12. Let $A = B = \mathbb{N}$ and $\varphi(m, n) = 1$ if $m \leq n$, $\varphi(m, n) = 0$ otherwise. Then φ has the order property and is therefore unstable. Here $\text{ltp}_\varphi(m)$ is the indicator function of $[m, \infty)$, and $\text{RS}(\varphi)$ consists of the indicator functions of the intervals $[0, n]$ together with the constant function **1**. Similarly, the space $\text{LS}(\varphi)$ consists of the functions

$\text{ltp}_\varphi(m)$ together with the constant function $\mathbf{0}$. So every $p \in \text{LS}(\varphi)$, viewed as a function on B , is either an instance $\varphi(m, \cdot)$ or the constant 0, which is trivially a polynomial in instances. Symmetrically, every $q \in \text{RS}(\varphi)$ is an instance or the constant 1. The failure appears at the unrealized pair $(\mathbf{0}, \mathbf{1})$. If $\hat{\varphi}$ were separately continuous and extended φ as in Theorem 1.2.11, then $\hat{\varphi}(\mathbf{0}, \text{rtp}_\varphi(n)) = \mathbf{0}(n) = 0$ for all n , and $\text{rtp}_\varphi(n) \rightarrow \mathbf{1}$, so $\hat{\varphi}(\mathbf{0}, \mathbf{1}) = 0$. But also $\hat{\varphi}(\text{ltp}_\varphi(m), \mathbf{1}) = \mathbf{1}(m) = 1$ for all m , and $\text{ltp}_\varphi(m) \rightarrow \mathbf{0}$, so $\hat{\varphi}(\mathbf{0}, \mathbf{1}) = 1$.

Corollary 1.2.13. *Let M be an L -structure and let $\varphi(x, y)$ be a formula that is stable on M . Then every $p \in \text{LS}(\varphi)$ is defined by a Boolean combination of instances: there are $a_1, \dots, a_k \in M^{|x|}$ and a Boolean combination $\psi(y)$ of $\varphi(a_1, y), \dots, \varphi(a_k, y)$ such that, for all $b \in M^{|y|}$, we have*

$$p(b) = 1 \iff M \models \psi(b).$$

Proof. Apply Theorem 1.2.11 with $\varepsilon = \frac{1}{3}$ to get $a_1, \dots, a_k$ and P . Both $p(b)$ and $\varphi(a_i, b)$ lie in $\{0, 1\}$, so $p(b) = 1$ if and only if $t = (\varphi(a_1, b), \dots, \varphi(a_k, b))$ belongs to $S = \{t \in \{0, 1\}^k : P(t) > \frac{1}{2}\}$. Take $\psi(y) = \bigvee_{t \in S} \bigwedge_{i \leq k} \varphi(a_i, y)^{t_i}$ (recall that $\varphi^1 = \varphi$ and $\varphi^0 = \neg\varphi$). ☺

The left types of a formula are its φ -types in another guise. Let M be a model of T , and identify each $p \in S_\varphi(M)$ with the characteristic function of D_p on $M^{|y|}$. With this identification we have $\text{LS}(\varphi) = S_\varphi(M)$. Indeed, for $a \in M$ the left type $\text{ltp}_\varphi(a)$ is $\text{tp}_\varphi(a/M)$, and by Lemma 1.1.2, every $p \in S_\varphi(M)$ agrees on each finite set of parameters with some realized type, so p is a pointwise limit of realized types. Conversely, a pointwise limit of realized types determines a set of φ -formulas over M that is finitely satisfiable in M , hence consistent with the elementary diagram of M , and so a complete φ -type.

In this two-valued setting, the quotient map $\beta M^{|x|} \rightarrow S_\varphi(M)$ is transparent. Recall from Section 1.1 that $S_\varphi(M)$ is the Stone space of the Boolean algebra $\mathcal{B}_\varphi(M)$ generated by the sets $\varphi(M, b)$ with $b \in M^{|y|}$. An ultrafilter $\mathcal{U}$ on $M^{|x|}$ goes to its trace on $\mathcal{B}_\varphi(M)$, that is, to the type containing $\varphi(x, b)$ exactly when $\varphi(M, b) \in \mathcal{U}$. The realized types are the traces of principal ultrafilters, and Lemma 1.1.2 says that every type is the trace of some ultrafilter.

Since a formula without the order property is stable on every model of T (see Remark 1.2.2), Corollary 1.2.13 already gives (1) $\Rightarrow$ (3) of Theorem 1.1.6, by analytic means. The price is positivity: the definitions obtained this way are Boolean combinations of instances rather than positive ones, and their size depends on p .



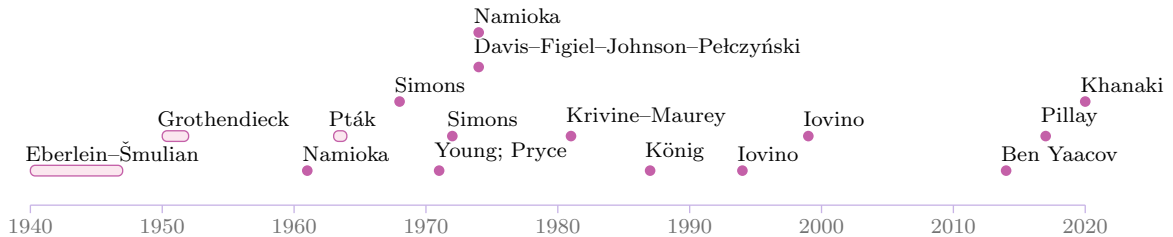
Remark 1.2.14. (1) *Lattice version.* The lattice form of the Stone–Weierstrass theorem replaces polynomials by the operations $\max$, $\min$ and affine maps. This is the form used in continuous logic, where the lattice form shows that a small set of connectives generates all continuous connectives up to uniform approximation [43, Proposition 1.4].

(2) *Convex combinations.* Stability gives more than a polynomial approximation. In Proposition A.1.7 below we show, using Grothendieck’s criterion, that stability of φ implies that $\{e_a : a \in A\}$ is relatively weakly compact in $C(\text{RS}(\varphi))$. In that case $p = \lim_{i, \mathcal{U}} \text{ltp}_\varphi(a_i)$ implies that $\hat{\varphi}(p, \cdot)$ is the weak limit of the functions e_{a_i} along $\mathcal{U}$. Mazur’s lemma then shows that $\hat{\varphi}(p, \cdot)$ is a uniform limit of convex combinations $\sum_j \lambda_j e_{a_{i_j}}$. So p is a uniform limit on B of averages of instances $\varphi(a, y)$. Ben Yaacov [42] uses this convex approximation to recover the familiar forms of the defining formulas.

1.3. Historical remarks

✂ **Section 1.1.** Counting types begins with Morley’s proof of Łoś’s conjecture [35]. There he introduced totally transcendental theories, defined by a restriction on the topology of the type spaces of those theories, and showed that for countable theories total transcendence is equivalent to a restriction on the number of types: only countably many types over countable sets, the condition now called ω -stability. The general notion arose soon after. Setting out to extend Morley’s categoricity theorem to uncountable theories, Rowbottom generalized ω -stability to λ -stability: a theory is λ -stable if there are at most λ types over sets of size λ [66]. Shelah then called a theory *stable* when λ -stable for some λ , and began the study of stable theories in general in [41]. In [36] he moved from theories to single formulas, defining when a formula $\varphi(x; y)$ is *stable in λ* : the formula φ has at most λ φ -types over sets of size λ [36, Definition 2.1]. He also defined the *order property* [36, Definition 2.2] and proved that stability is equivalent to the absence of the order property. The ladder formulation of Definition 1.1.4 is a finitary restatement of Shelah’s condition. Definable types were introduced by Gaifman in his study of end extensions of models of Peano arithmetic [40], and Shelah arrived at them independently. The theorem that φ -types are definable when φ is stable is [36, Theorem 3.1]; see also [57, Chapter II]. Today this theorem is often called the fundamental theorem of stability theory. Ben Yaacov [42] observed that the same result follows from Grothendieck’s 1952 work on weak compactness in function spaces [44] (see Appendix A below). This is the viewpoint of Section 1.2.

✂ **Section 1.2.1.** Interchanging iterated limits is an old theme. For double sequences of reals, the classical sufficient condition is uniform convergence in one variable (the Moore–Osgood theorem). The condition of Definition 1.2.5 requires the iterated limits to agree along *all* pairs of sequences. This condition comes from functional analysis, and reached model theory through Banach space geometry.



1940–1947

Eberlein–Šmulian. Šmulian (1940) and Eberlein [17] (1947) prove that in a Banach space weak compactness and weak sequential compactness coincide, so that weak compactness can be tested along sequences.

1950–1952

Grothendieck. Grothendieck proves the *double limit criterion*, announced in [18] and proved in [44]. For compact K , a bounded set $H \subseteq C(K)$ is relatively weakly compact if and only if $\lim_m \lim_n f_n(x_m) = \lim_n \lim_m f_n(x_m)$ for all sequences (f_n) in H and (x_m) in K for which both iterated limits exist. For an arbitrary topological space, the same condition decides when pointwise limits of functions in H are continuous. See Appendix A.

1961

Namioka. Namioka proves a combinatorial “substitute for Lebesgue’s bounded convergence theorem” [23]. From this result he obtains, without measure

theory, Krein's theorem: in a Banach space, the closed convex hull of a weakly compact set is weakly compact.

1963–1964

Pták. Pták [26] finds a purely combinatorial lemma about averages, from which both the double limit criterion and Krein's theorem follow. In [27] he uses an interchange of iterated limits to decide which bounded functions on a product $X \times Y$, continuous in each variable separately, extend to bilinear forms with the same kind of continuity.

1968

Simons. Simons [28] introduces the *Fubini characterization* of the interchange of double limits. A bounded function on a product satisfies the iterated limit condition if and only if the two iterated integrals of the function agree, for every pair of measures on the factors. This characterization plays an important role in the proof of the factorization theorem of Davis, Figiel, Johnson and Pełczyński [46].

1971

Young; Pryce. N. J. Young [32] gives topological proofs of Pták's double-limit theorems, based on Grothendieck's work rather than combinatorics, and generalizes Pták's bilinear extension theorem. Pryce [25] uses a trick of Whitley's to simplify proofs about pointwise compact sets of continuous functions.

1972

Simons. In three consecutive papers, Simons gives an elementary proof of Pták's lemma [31]. He also proves an inequality about suprema and upper limits, now known as *Simons' inequality* [29], and relates minimax theorems, which say when $\max \min = \min \max$, to James's characterization of reflexivity [30].

1974

Davis–Figiel–Johnson–Pełczyński. Davis, Figiel, Johnson and Pełczyński [46] prove that every weakly compact operator factors through a reflexive Banach space. This is the key step in the proof of Theorem 1.2.9.

1974

Namioka. Namioka [24] studies when separately continuous functions on products are jointly continuous at many points. This is the question behind Proposition 1.2.8.

1981

Krivine–Maurey. Krivine and Maurey [51] apply the interchange of limits to the norm: a Banach space is *stable* if $\lim_m \lim_n \|x_m + y_n\| = \lim_n \lim_m \|x_m + y_n\|$ for all bounded sequences for which both limits exist. They prove that every stable Banach space contains, for some p , copies of ℓ_p that are as close to isometric as desired, which generalizes a theorem of Aldous about subspaces of L_1 .

1987

König. König [22] studies the double limit relation in angelic spaces, where compactness can be tested by sequences.

1994

Iovino. Iovino's Ph.D. thesis [47] introduces stability for real-valued logic.

- 1999

Iovino. Iovino [49] observes that Shelah’s order property is the 2-valued case of the same condition, and proves Theorem 1.2.9. Propositions 1.2.4 and 1.2.3 are taken from his paper.
 - 2014

Ben Yaacov. Ben Yaacov [42] derives the definability of types in stable theories directly from Grothendieck’s criterion.
 - 2017

Pillay. Pillay [21] comments on the model-theoretic reading of Grothendieck’s criterion, and derives from the criterion the positive form of the definitions.
 - 2020

Khanaki. Khanaki [20] studies stability, NIP (see Chapter 5), and related properties systematically, through topological properties of function spaces.
- ⌘ **Section 1.2.2.** Spaces of left and right φ -types (see Definition 1.2.7) are the analytic counterpart of Shelah’s spaces of φ -types. The use of *types* in Banach space theory, as limits of functions $y \mapsto \|x_n + y\|$, is due to Krivine and Maurey [51]. Proposition 1.2.8, which characterizes stability by the existence of a separately continuous extension to pairs of types, is Proposition 2.6 of Iovino [49]. Separately continuous functions on products of compact spaces, and their relation to iterated limits, are the subject of Grothendieck’s paper [44].
- ⌘ **Section 1.2.3.** James [50] characterized reflexivity by the absence of sequences (a_m) in B_E and (f_n) in B_{E^*} whose pairing takes one value for $m < n$ and another for $m \geq n$. This is a Banach-space form of the order property. In the early 1980s, Krivine and Maurey [51] and Raynaud [52] showed that stable functions give rise to weakly compact operators. The factorization theorem of Davis, Figiel, Johnson and Pełczyński [46] (1974) says that every weakly compact operator factors through a reflexive space. Iovino [49] combined these results to prove Theorem 1.2.9: stable formulas are exactly the pairings between the unit balls of a reflexive space E and of E^* .
- ⌘ **Section 1.2.4.** Weierstrass proved in 1885 that every continuous function on a compact interval is a uniform limit of polynomials [39]. Stone replaced the interval by an arbitrary compact Hausdorff space and the coordinate function by any family of functions that separates points [37, 38]. Ben Yaacov and Usvyatsov [43] used Stone’s theorem to lay the foundations of continuous first-order logic. There Stone’s theorem shows that finitely many connectives suffice and that continuous functions on type spaces are exactly the definable predicates. Ben Yaacov [42] combined Grothendieck’s criterion with Mazur’s lemma [34] (a weakly convergent sequence has convex combinations converging in norm), and so obtained the defining formulas of stable types as uniform limits of averages of instances; see Remark 1.2.14.

Chapter 2

Stability, trees, and definability of types

This chapter proves the definability theorem, Theorem 1.1.6, which Chapter 1 stated without proof. The argument passes through binary trees, and along the way we show that a formula is stable exactly when it cannot encode tall trees. As in Section 1.1, we fix a first-order theory T in a language L and an L -formula $\varphi(x, y)$ consistent with T .

2.1. The equivalence proved in this chapter

Four conditions on φ turn out to coincide: instability, the order property, the presence of tall binary trees, and the existence of a non-definable φ -type. The notions of φ -type, definable type, ladder, order property and stability come from Section 1.1; the newcomer, a binary tree for φ , needs a little notation for finite 0–1 sequences.

Notation 2.1.1. Recall that $\varphi^1 = \varphi$ and $\varphi^0 = \neg\varphi$. For an ordinal μ , we write $2^{<\mu}$ for the set of 0–1 sequences of length $<\mu$, and we write $2^{\leq n} = 2^{<n} \cup 2^n$. We write $\sigma \sqsubseteq \tau$ if σ is an initial segment of τ , $\sigma \sqsubset \tau$ if moreover $\sigma \neq \tau$, $|\sigma|$ for the length of σ , $\sigma \frown \langle u \rangle$ for the extension of σ by one entry u , and, for η of length $>k$, $\eta \upharpoonright k$ for the initial segment of η of length k .

Definition 2.1.2. A *tree of height n for φ in a model M of T* consists of families $(b_\sigma : \sigma \in 2^{<n})$ and $(a_\eta : \eta \in 2^n)$ of tuples of M such that

$$\sigma \frown \langle u \rangle \sqsubseteq \eta \implies M \models \varphi(a_\eta, b_\sigma)^u.$$

In other words, the leaf a_η satisfies $\varphi(x, b_\sigma)$ exactly when the branch η turns right ($u = 1$) at the node σ . We say that φ *encodes binary trees of arbitrarily large finite height* if for every n some model of T contains a tree of height n for φ .

Figure 2.1 shows a tree of height 3. Each node b_σ splits the leaves above it into two halves: the leaves on the right satisfy $\varphi(x, b_\sigma)$, and the leaves on the left satisfy $\neg\varphi(x, b_\sigma)$. The branch leading to a leaf a_η is thus a record, node by node, of which of the formulas $\varphi(x, b_\sigma)$ the leaf satisfies, and a tree of height n realizes all 2^n such records.

Like ladders, trees can be shortened. A tree of height $n + 1$ restricts to one of height n : keep the nodes b_σ with $\sigma \in 2^{<n}$, and for each $\eta \in 2^n$ keep the leaf $a_{\eta \frown \langle 0 \rangle}$.

Theorem 2.1.3. *Let $\varphi(x, y)$ be an L -formula. The following are equivalent.*

- (1) *The formula φ is unstable (see Definition 1.1.4).*
- (2) *The formula φ has the order property (see Definition 1.1.4).*
- (3) *The formula φ encodes binary trees of arbitrarily large finite height.*
- (4) *Some φ -type over some model of T is not definable (see Definitions 1.1.1 and 1.1.3).*

The proof goes around a cycle of implications. The equivalence (1) $\Leftrightarrow$ (2) is the definition of stability (see Definition 1.1.4). In Section 2.3 we turn long ladders into tall trees, which gives (2) $\Rightarrow$ (3) (see Proposition 2.3.1). In Section 2.4 we use tall trees to produce more φ -types than there are formulas to define them, which gives (3) $\Rightarrow$ (4) (see Proposition 2.4.2). The cycle closes with the definability theorem, proved in Section 2.2, whose contrapositive is (4) $\Rightarrow$ (2) (see Theorem 2.2.1 and Corollary 2.2.2).

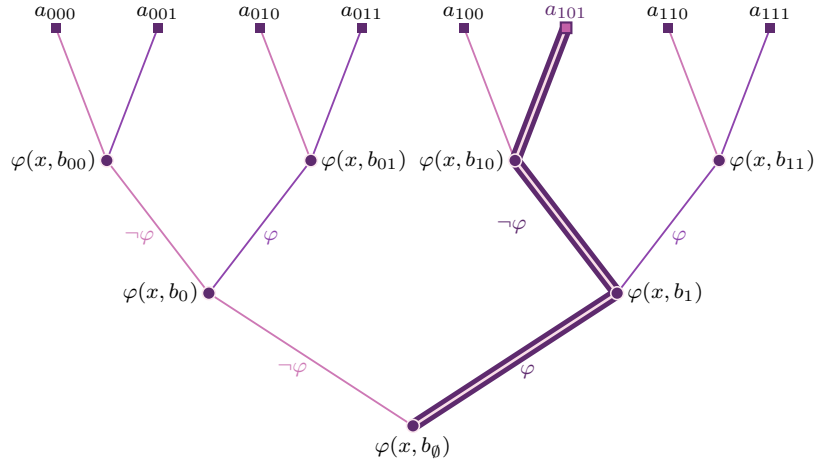


FIGURE 2.1. A tree of height 3 for φ . Each round node is labeled by the formula $\varphi(x, b_\sigma)$ given by its parameter b_σ , and the squares a_η are the leaves. An edge going up and to the right from the node $\varphi(x, b_\sigma)$ is labeled φ (purple), and an edge going up and to the left is labeled $\neg\varphi$ (pink): every leaf above a right edge out of that node satisfies $\varphi(x, b_\sigma)$, and every leaf above a left edge satisfies $\neg\varphi(x, b_\sigma)$. The edges into the leaves are not labeled, but their colors follow the same rule. For example, the thick branch leading up to a_{101} goes right, left, right, so $M \models \varphi(a_{101}, b_\emptyset) \wedge \neg\varphi(a_{101}, b_1) \wedge \varphi(a_{101}, b_{10})$. The formula φ is not required to say anything about a_η and b_σ when σ is not an initial segment of η , such as a_{101} and b_0 .

Chapter 3 gives alternative proofs of two of these implications, each of which yields information that the proofs in this chapter do not provide. Here the way from tall trees back to the order property passes through non-definable types, and so requires counting types over models of large cardinality. In Section 3.1 we prove $(3) \Rightarrow (2)$ directly instead, by finite combinatorics based on Ramsey's theorem (see Proposition 3.1.5 and Corollary 3.1.6). This direct proof stays inside a single model and is quantitative: a tree of height $2^{\binom{2n}{n}} - 1$ in a model M already yields an n -ladder in M . In Section 3.2 the classical rank argument proves $(4) \Rightarrow (3)$, again in contrapositive form: Proposition 3.2.3 shows that the absence of tall trees alone makes every φ -type definable, without going through the order property, and gives the definition explicitly in terms of a rank. Unlike the definitions of Section 2.2, however, these definitions are not positive Boolean combinations of instances of φ .

2.2. From the absence of the order property to type definability

In this section we prove $(1) \Rightarrow (2)$ of Theorem 1.1.6: if φ does not have the order property, then every φ -type over a model has a definition built from instances of φ using only $\wedge$ and $\vee$. The candidate definitions are easy to write down, and the work lies in choosing the finitely many parameters that they use.

Let M be a model of T , let $p \in S_\varphi(M)$ be given, and recall that $D_p = \{b \in M : \varphi(x, b) \in p\}$. For a finite set $F \subseteq M$, we shall write

$$\mathcal{W}_F = \{ \{a \in F : M \models \varphi(a, b)\} : b \in D_p \} \quad \text{and} \quad d_F \varphi(y) = \bigvee_{W \in \mathcal{W}_F} \bigwedge_{a \in W} \varphi(a, y).$$

The family $\mathcal{W}_F$ consists of subsets of the finite set F , so the family is finite and the formula $d_F\varphi$ is a positive Boolean combination of instances of φ over M . In words, the formula $d_F\varphi(y)$ says that, for some $b' \in D_p$, every $a \in F$ with $M \models \varphi(a, b')$ also satisfies $\varphi(a, y)$. Every parameter $b \in D_p$ satisfies $d_F\varphi$, whatever F is (take $b' = b$). The task is to choose F so that no parameter outside D_p satisfies $d_F\varphi$, and this is where the absence of the order property is used.

Theorem 2.2.1. *Assume that $\varphi(x, y)$ does not have the order property, let M be a model of T , and let $p \in S_\varphi(M)$ be given. Then there is a finite set $F \subseteq M$ such that $d_F\varphi$ is a φ -definition of p ; that is, for all $b \in M$ we have $\varphi(x, b) \in p$ if and only if $M \models d_F\varphi(b)$.*

Proof. For $b \in M$ we shall write $S_b = \{a \in M : M \models \varphi(a, b)\}$. The only property of p we use is that p is finitely satisfiable in M (see Lemma 1.1.2).

© **Claim.** There is a finite $F \subseteq M$ such that $S_b \cap F \not\subseteq S_{b'}$ for all $b \in D_p$ and $b' \in M \setminus D_p$.

Granting the claim, we check that $d_F\varphi$ defines p . Let $b \in M$ be given.

- (a) If $b \in D_p$, then $W := S_b \cap F \in \mathcal{W}_F$ and $W \subseteq S_b$, so $M \models \bigwedge_{a \in W} \varphi(a, b)$. Hence $M \models d_F\varphi(b)$.
- (b) If $b \notin D_p$ and $W \in \mathcal{W}_F$, say $W = S_{b_0} \cap F$ with $b_0 \in D_p$, then $W \not\subseteq S_b$ by the claim. So some $a \in W$ has $M \models \neg\varphi(a, b)$, and $M \models \neg\bigwedge_{a \in W} \varphi(a, b)$. As this holds for every $W \in \mathcal{W}_F$, we conclude that $M \models \neg d_F\varphi(b)$.

© **Proof of the claim.** Suppose the claim fails. Then for every finite $F \subseteq M$ there are $b \in D_p$ and $b' \in M \setminus D_p$ with $S_b \cap F \subseteq S_{b'}$, i.e. $M \models \varphi(a, b) \rightarrow \varphi(a, b')$ for all $a \in F$. Letting the failure of the claim and the finite satisfiability of p take turns, we build a configuration in which Ramsey's theorem will find a ladder. We choose recursively, for $i < \omega$, elements $b_i \in D_p$, $b'_i \in M \setminus D_p$ and $a_i \in M$ as follows.

- (a) Apply the failure of the claim to $F = \{a_j : j < i\}$ to get $b_i \in D_p$ and $b'_i \in M \setminus D_p$ with

$$(2.1) \quad M \models \varphi(a_j, b_i) \rightarrow \varphi(a_j, b'_i) \quad \text{for all } j < i.$$

- (b) Since $\varphi(x, b_k) \in p$ and $\neg\varphi(x, b'_k) \in p$ for $k \leq i$, Lemma 1.1.2 gives $a_i \in M$ with

$$(2.2) \quad M \models \varphi(a_i, b_k) \wedge \neg\varphi(a_i, b'_k) \quad \text{for all } k \leq i.$$

Color each pair $j < i$ by the pair of truth values $(\varphi(a_j, b_i), \varphi(a_j, b'_i)) \in \{0, 1\}^2$. By (2.1), the color $(1, 0)$ never occurs. By Ramsey's theorem there is an infinite $H = \{h_0 < h_1 < \dots\} \subseteq \omega$ on which all pairs have the same color c . Let $s, t < \omega$ be given.

© **Case $c \in \{(0, 1), (1, 1)\}$.** If $s < t$, then $M \models \varphi(a_{h_s}, b'_{h_t})$ by homogeneity. If $s \geq t$, then $M \models \neg\varphi(a_{h_s}, b'_{h_t})$ by (2.2). So $\varphi(a_{h_s}, b'_{h_t})$ holds if and only if $s < t$, and with $\beta_t = b'_{h_{t+1}}$ we get $M \models \varphi(a_{h_s}, \beta_t) \iff s \leq t$.

© **Case $c = (0, 0)$.** If $s < t$, then $M \models \neg\varphi(a_{h_s}, b_{h_t})$ by homogeneity. If $s \geq t$, then $M \models \varphi(a_{h_s}, b_{h_t})$ by (2.2). So $\varphi(a_{h_s}, b_{h_t})$ holds if and only if $s \geq t$. For a given n , reversing the first n indices by putting $\alpha_s = a_{h_{n-1-s}}$ and $\beta_t = b_{h_{n-1-t}}$ gives $M \models \varphi(\alpha_s, \beta_t) \iff s \leq t$ for $s, t < n$.

In both cases M contains an n -ladder for every n , so φ has the order property, a contradiction. This proves the claim and the theorem. $\odot$

Corollary 2.2.2. *If φ does not have the order property, then for every model M of T , every $p \in S_\varphi(M)$ is definable by a positive Boolean combination of instances $\varphi(a, y)$ with $a \in M$. In particular, if φ is quantifier-free, then p has a quantifier-free φ -definition.*

The proof of Theorem 2.2.1 applies Ramsey's theorem for pairs to an infinite sequence, inside an argument by contradiction, and so says nothing about sizes. A finite Ramsey-type argument, adapted to binary trees rather than to sequences, reappears in Section 3.1, where it extracts ladders of prescribed length directly from trees of explicit height.

2.3. From the order property to tall trees

To prove (3) $\Rightarrow$ (1) of Theorem 1.1.6, we need a supply of many φ -types over a small model, and binary trees (see Definition 2.1.2) are the natural source. The first step, taken in this section, is to grow tall trees out of long ladders. The idea is to place the nodes and the leaves of a binary tree on a line, in such a way that the leaves above the right edge out of each node lie to the left of that node and the leaves above the left edge lie to the right of the node. A ladder read along this line then has exactly the pattern of a tree.

Proposition 2.3.1. *If a model M of T contains an N -ladder for φ , where $N = 2^{n+1} - 1$, then M contains a tree of height n . Hence, if φ has the order property, then φ encodes binary trees of arbitrarily large finite height.*

Proof. For $\sigma \in 2^{\leq n}$ put

$$d(\sigma) = \sum_{k < |\sigma|} (1 - \sigma(k)) 2^{-(k+1)} + 2^{-(|\sigma|+1)} \in (0, 1).$$

The binary expansion of $d(\sigma)$ is $(1 - \sigma(0), \dots, 1 - \sigma(|\sigma| - 1), 1)$, which ends in 1. So d is injective on $2^{\leq n}$, and the image V of d has exactly $|2^{\leq n}| = 2^{n+1} - 1 = N$ elements.

© **Claim.** If $\sigma \frown \langle u \rangle \subseteq \eta$ with $\eta \in 2^n$, then $d(\eta) < d(\sigma) \iff u = 1$.

Indeed, let $k = |\sigma|$. Since $\eta|k = \sigma$ and $\eta(k) = u$, we have

$$d(\eta) - d(\sigma) = -u 2^{-(k+1)} + R, \quad R = \sum_{k < j < n} (1 - \eta(j)) 2^{-(j+1)} + 2^{-(n+1)}.$$

Here $0 < R \leq (2^{-(k+1)} - 2^{-n}) + 2^{-(n+1)} < 2^{-(k+1)}$. Hence $d(\eta) - d(\sigma)$ is negative if $u = 1$ and positive if $u = 0$, which proves the claim.

Let $a_1, \dots, a_N, b_1, \dots, b_N \in M$ satisfy $\varphi(a_i, b_j) \iff i \leq j$, and let $e: V \rightarrow \{1, \dots, N\}$ be the order isomorphism. Put $b_\sigma := b_{e(d(\sigma))}$ for $\sigma \in 2^{\leq n}$ and $a_\eta := a_{e(d(\eta))}$ for $\eta \in 2^n$. If $\sigma \frown \langle u \rangle \subseteq \eta$, then $d(\eta) \neq d(\sigma)$, and so

$$M \models \varphi(a_\eta, b_\sigma) \iff e(d(\eta)) \leq e(d(\sigma)) \iff d(\eta) < d(\sigma) \iff u = 1.$$

Thus $(b_\sigma), (a_\eta)$ is a tree of height n in M . ☺

Figure 2.2 shows the construction for $n = 2$.

2.4. Tall trees produce non-definable types

Tall trees produce more φ -types over a small model than there are formulas to define them, and this proves (3) $\Rightarrow$ (4) of Theorem 2.1.3. The count is done in Proposition 2.4.2 below. To prepare for it, in Lemma 2.4.1 below we use compactness to turn finite trees into a tree of infinite height, whose branches will give the types. When L is countable, the count is easiest to see over a countable model, and this case is the picture to keep in mind: a tree of height ω has $2^{\aleph_0}$ branches, each branch gives a φ -type over a countable model, and only countably many formulas are available to define them. In Section 2.5 the same tree of height ω reappears as a copy of the Cantor space inside $S_\varphi(M)$.

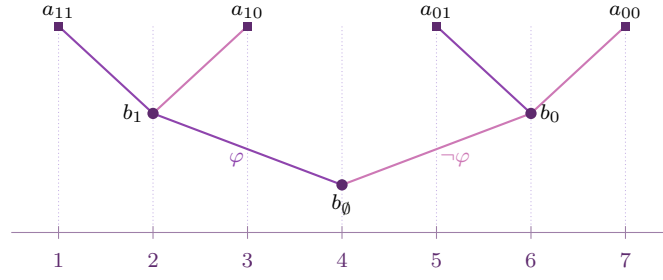


FIGURE 2.2. The proof of Proposition 2.3.1 for $n = 2$. The seven sequences of length at most 2 are placed at the positions $e(d(\sigma)) = 1, \dots, 7$ of a ladder of length 7, and the tree grows upward. In a ladder, $\varphi(a_i, b_j)$ holds when a_i lies to the left of b_j (that is, $i \leq j$), so here the φ -edges (purple) go up and to the left and the $\neg\varphi$ -edges (pink) go up and to the right. The bound $R < 2^{-(k+1)}$ in the claim is what keeps every leaf on the correct side of each node below it: the leaf a_{10} , for instance, lies to the left of $b_\emptyset$ and to the right of b_1 .

Lemma 2.4.1 (compactness). *Suppose that φ encodes binary trees of arbitrarily large finite height, and let μ be an infinite cardinal. Then there are a model M' of T and tuples $(b_s)_{s \in 2^{<\mu}}$ in M' such that for every $\eta \in 2^\mu$ the set*

$$\Sigma_\eta(x) = \{\varphi(x, b_{\eta|_\alpha})^{\eta(\alpha)} : \alpha < \mu\}$$

is finitely satisfiable in M' .

Proof. Add new tuples of constants c_s ($s \in 2^{<\mu}$) and let Γ be T together with the sentences

$$\gamma_{\eta, F} := \exists x \bigwedge_{\alpha \in F} \varphi(x, c_{\eta|_\alpha})^{\eta(\alpha)} \quad (\eta \in 2^\mu, F \subseteq \mu \text{ finite}).$$

By compactness, we need only show that every finite $\Gamma_0 \subseteq \Gamma$ has a model.

Let γ_{η_i, F_i} ($i < m$) be the sentences of this form in Γ_0 (if $m = 0$ there is nothing to do). Only finitely many nodes of $2^{<\mu}$ are mentioned in these sentences, and the plan is to embed them into a finite tree, preserving the direction in which each branch leaves each node. Let S be the finite set of all $\eta_i|_\alpha$ and $\eta_i|(\alpha + 1)$ with $i < m$ and $\alpha \in F_i$. Note that $\alpha + 1 < \mu$ because μ is a limit ordinal. Let $S' = \{s \wedge s' : s, s' \in S\}$, where $s \wedge s'$ is the longest common initial segment. The set S' is finite and closed under $\wedge$, because in a tree $s \wedge s' \wedge s''$ equals one of the pairwise meets. So S' has a least element r , and every $s \in S' \setminus \{r\}$ has a longest predecessor $s^- \in S'$ with $s^- \sqsubset s$.

Define $f: S' \rightarrow 2^{<N}$, where $N = |S'| + 1$, by recursion on length, setting $f(r) = \emptyset$ and $f(s) = f(s^-) \frown \langle s(|s|) \rangle$. Then $|f(s)| < N$.

⊙ **Claim.** If $s, s' \in S'$ and $s \sqsubset s'$, then $f(s) \frown \langle s'(|s|) \rangle \sqsubseteq f(s')$.

The proof is by induction on $|s'|$. If $s'^- = s$, this is the definition of f . Otherwise $s \sqsubset s'^- \sqsubset s'$ by maximality of s'^- . By induction, we have $f(s) \frown \langle s'^-(|s|) \rangle \sqsubseteq f(s'^-) \sqsubseteq f(s')$, and finally $s'^-(|s|) = s'(|s|)$.

By hypothesis, take a model M_0 of T containing a tree $(b_\sigma)_{\sigma \in 2^{<N}}$, $(a_\nu)_{\nu \in 2^N}$ of height N . Interpret c_s as $b_{f(s)}$ for $s \in S'$, and arbitrarily otherwise. Fix $i < m$, let $\alpha^* = \max F_i$ and $s^* = \eta_i|(\alpha^* + 1) \in S'$. Pick $\nu \in 2^N$ extending $f(s^*)$. For $\alpha \in F_i$ we have $\eta_i|_\alpha \sqsubset s^*$ and $s^*(\alpha) = \eta_i(\alpha)$. By the Claim, we have $f(\eta_i|_\alpha) \frown \langle \eta_i(\alpha) \rangle \sqsubseteq \nu$, and so $M_0 \models \varphi(a_\nu, b_{f(\eta_i|_\alpha)})^{\eta_i(\alpha)}$. Hence a_ν witnesses γ_{η_i, F_i} , and M_0 with this interpretation is a model of Γ_0 .

By compactness Γ has a model. The L -reduct M' of that model, with $b_s = c_s^{M'}$, is as required. ☺

In the next proposition we write $\kappa = |L| + \aleph_0$, which is the number of L -formulas.

Proposition 2.4.2. *If φ encodes binary trees of arbitrarily large finite height, then for every cardinal $\lambda \geq \kappa$ there is a model M of T with $|M| \leq \lambda$ and $|S_\varphi(M)| > \lambda$. In particular, there are a model M of T with $|M| \leq \kappa$ and a type $p \in S_\varphi(M)$ such that p is not definable.*

Proof. Let μ be the least cardinal with $2^\mu > \lambda$. Then μ is infinite and $\mu \leq \lambda$. For $\nu < \mu$ we have $2^\nu \leq \lambda$, so $|2^{<\mu}| = \sum_{\nu < \mu} 2^\nu \leq \mu \cdot \lambda = \lambda$.

Take M' and (b_s) as in Lemma 2.4.1. By the downward Löwenheim–Skolem theorem, there is $M \preceq M'$ with $|M| \leq \lambda$ containing all b_s . Each finite part of Σ_η is satisfiable in M' , hence in M , since $\exists x \bigwedge \dots$ is an $L(M)$ -sentence. So Σ_η is consistent with the elementary diagram of M and extends to some $p_\eta \in S_\varphi(M)$.

If $\eta \neq \eta'$, let α be least with $\eta(\alpha) \neq \eta'(\alpha)$. Then $s = \eta|_\alpha = \eta'|_\alpha$, and p_η and $p_{\eta'}$ disagree on $\varphi(x, b_s)$. Hence $|S_\varphi(M)| \geq 2^\mu > \lambda$.

For the last statement take $\lambda = \kappa$. A definable type is determined by any φ -definition of the type, and there are only $|L(M)| \leq \kappa$ formulas over M . So at most κ types in $S_\varphi(M)$ are definable, and some $p \in S_\varphi(M)$ is not. ☺

Proof of (3) $\Rightarrow$ (1) in Theorem 1.1.6. If φ has the order property, then φ encodes binary trees of arbitrarily large finite height by Proposition 2.3.1, and so some φ -type over some model of T is not definable by Proposition 2.4.2. ☺

This completes the proof of Theorem 1.1.6, and the proof leaves behind a characterization of the order property by trees.

Theorem 2.4.3. *The formula φ has the order property if and only if φ encodes binary trees of arbitrarily large finite height.*

Proof. The forward direction is Proposition 2.3.1. Conversely, if φ encodes tall trees, then some φ -type is not definable (see Proposition 2.4.2), so φ has the order property by (1) $\Rightarrow$ (3) of Theorem 1.1.6. ☺

2.5. Type spaces over countable models: a perfect set dichotomy

Section 2.4 measured instability by counting types; topology measures instability by shape, and over countable models the two measurements agree. Throughout this section the language L is countable. For a model M of T , the space $S_\varphi(M)$ is the Stone space of the Boolean algebra $\mathcal{B}_\varphi(M)$ (see Section 1.1). If M is countable, the algebra $\mathcal{B}_\varphi(M)$ is countable, so its Stone space $S_\varphi(M)$ is a compact metrizable zero-dimensional space. By the Cantor–Bendixson theorem, such a space is either countable, in which case the space is scattered, or contains a homeomorphic copy of the Cantor space 2^ω . Stability decides which alternative occurs.

Theorem 2.5.1. *Assume that L is countable. The following are equivalent.*

- (1) *The formula φ is stable.*
- (2) *For every countable model M of T , the space $S_\varphi(M)$ is countable.*
- (3) *For every countable model M of T , the space $S_\varphi(M)$ is scattered.*

Moreover, if φ is unstable, then some countable model M of T has a closed subset of $S_\varphi(M)$ homeomorphic to 2^ω .

Proof. (1) $\Rightarrow$ (2). By Theorem 1.1.6, every $p \in S_\varphi(M)$ has a φ -definition, which is an $L(M)$ -formula, and the definition determines p . Since there are only countably many $L(M)$ -formulas, the space $S_\varphi(M)$ is countable.

(2) $\Leftrightarrow$ (3). This is the Cantor–Bendixson theorem for compact metrizable spaces.

(2) $\Rightarrow$ (1) and the last statement. Suppose that φ is unstable. By Theorem 2.4.3, the formula φ encodes binary trees of arbitrarily large finite height, so Lemma 2.4.1 with $\mu = \omega$ gives a model M' of T and tuples $(b_\sigma)_{\sigma \in 2^{<\omega}}$ in M' such that every set $\Sigma_\eta(x)$, for $\eta \in 2^\omega$, is finitely satisfiable in M' . By the downward Löwenheim–Skolem theorem, there is a countable model $M \preceq M'$ containing every b_σ , and by elementarity every $\Sigma_\eta(x)$ is finitely satisfiable in M . For $\sigma \in 2^{<\omega}$, let C_σ be the clopen set of types $p \in S_\varphi(M)$ that contain $\varphi(x, b_{\sigma|\alpha})^{\sigma(\alpha)}$ for every $\alpha < |\sigma|$. Each C_σ is nonempty, because the formulas in question form a finitely satisfiable subset of Σ_η for any η extending σ , and C_σ is the disjoint union of $C_{\sigma \smallfrown \langle 0 \rangle}$ and $C_{\sigma \smallfrown \langle 1 \rangle}$, which disagree on $\varphi(x, b_\sigma)$. So the sets C_σ form a Cantor scheme. The closed set $D = \bigcap_n \bigcup_{|\sigma|=n} C_\sigma$ maps continuously onto 2^ω , by sending p to the unique η with $p \in C_{\eta|n}$ for every n ; the map is onto because each $\bigcap_n C_{\eta|n}$ is a decreasing intersection of nonempty closed sets in a compact space. Hence $S_\varphi(M)$ is uncountable, so (2) fails, and by the Cantor–Bendixson theorem the uncountable compact metrizable space $S_\varphi(M)$ contains a closed copy of 2^ω . ☺

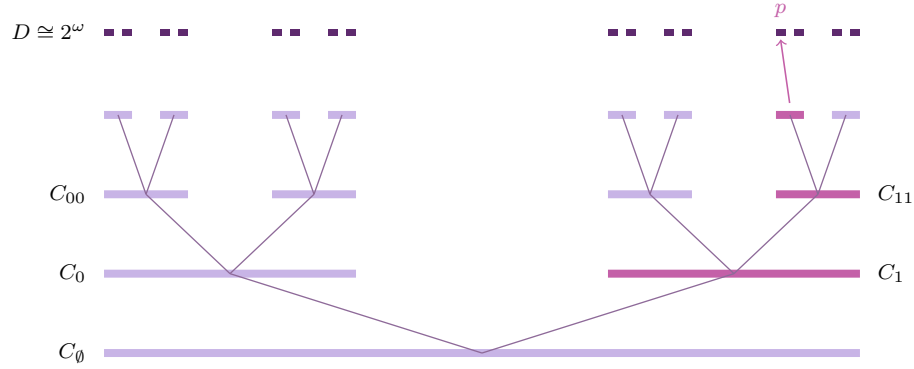


FIGURE 2.3. The Cantor scheme in the proof of Theorem 2.5.1, growing upward. Each bar is a nonempty clopen set C_σ of φ -types, and C_σ splits into $C_{\sigma \smallfrown \langle 0 \rangle}$ (left) and $C_{\sigma \smallfrown \langle 1 \rangle}$ (right), which disagree on $\varphi(x, b_\sigma)$. The pink bars $C_1 \supseteq C_{11} \supseteq C_{110} \supseteq \dots$ follow one branch η , and their intersection contains a type p with $p \in C_{\eta|n}$ for every n . Distinct branches give distinct types, so the types along the $2^{\aleph_0}$ branches form the closed set D , which maps onto 2^ω .

A tree of infinite height is thus nothing but a Cantor scheme of clopen sets in a type space (see Figure 2.3), and stability says that type spaces over countable models are scattered. This is the same topology that Morley used to define totally transcendental theories, whose type spaces over countable sets are scattered, and whose Morley rank is the Cantor–Bendixson rank on type spaces over $\aleph_0$ -saturated models. The rank R of Section 3.2 is a binary splitting rank on the clopen sets $[\theta]$, a finite combinatorial cousin of the Cantor–Bendixson rank: Lemma 3.2.2(iii) says that splitting a clopen set into two pieces of rank at least m raises the rank.

2.6. Historical remarks

- ⌘ **Section 2.2.** Definability of φ -types over models for formulas without the order property is due to Shelah [36, 57]. The sharper form proved here, in which the definition is a positive Boolean combination of instances of φ itself, goes back to Hrushovski and Pillay [19], as Pillay points out in [56, Chapter 1]. Pillay [21, Remark 2.4] derives the positive form directly from Grothendieck’s double limit criterion (see Theorem A.1.3 below) by a compactness argument. Positivity matters in applications: for example, positivity gives quantifier-free definitions for quantifier-free formulas.
- ⌘ **Section 2.3.** Binary trees enter stability theory through Shelah’s local rank $R(-, \varphi, 2)$. In [36, Lemma 2.7] the rank of a type is computed by the consistency of the family of formulas indexed by the branches of a binary tree of height n . That is, the formulas of infinite 2-rank are exactly those that encode binary trees of arbitrarily large finite height (see Definition 2.1.2). Shelah showed that this happens exactly when φ has the order property [36, 57]. The two directions behave very differently quantitatively. A ladder of length exponential in n yields a tree of height n (see Proposition 2.3.1). In the other direction, a tree yields only a ladder of length logarithmic in the height of the tree; see Section 3.1.
- ⌘ **Section 2.4.** The construction of this section is the local form of Shelah’s counting argument [36]. A tree of infinite height gives $2^{\aleph_0}$ φ -types over a countable set of parameters, which is incompatible with definability of all φ -types. This is also the origin of the *stability spectrum*: a theory with the order property has 2^λ types over some set of size λ , for every infinite λ [57].

Chapter 3

More stability, trees, and definability

Chapter 2 established the equivalences of Theorem 2.1.3 by the most direct route to the definability theorem. In that route, the way from tall trees back to the order property goes through non-definable types and a count of types over large models. This chapter develops the quantitative side of the theory, which that route leaves out. In Section 3.1 we show by a direct combinatorial argument, based on Ramsey’s theorem and carried out inside a single model, how tall a tree must be to contain a ladder of a given length. In Section 3.2 we present the classical proof of definability of types, which measures types by the height of the trees that can be grown above them and produces the definitions explicitly from this rank, without passing through the order property. The conventions are those of Chapter 2.

3.1. A quantitative combinatorial proof (using Ramsey’s theorem) that trees give the order property

Theorem 2.4.3 was derived from the definability theorem; here we give a direct and quantitative proof of the implication “tall trees $\Rightarrow$ order property”. The proof is pure finite combinatorics, the Boolean case of a Ramsey-type argument due to Hodges, and we follow the presentation of Conant and Terry [45]. Finding a ladder inside a tree requires passing to smaller trees embedded in it, and the first definition says what such an embedding is.

Definition 3.1.1. Let $s, t \geq 1$ be given.

- (a) A *node embedding* is a map $\theta: 2^{<s} \rightarrow 2^{<t}$ such that $\rho \frown \langle u \rangle \sqsubseteq \tau$ implies $\theta(\rho) \frown \langle u \rangle \sqsubseteq \theta(\tau)$.
- (b) A *full embedding* is a map $\theta: 2^{\leq s} \rightarrow 2^{\leq t}$ sending $2^{<s}$ into $2^{<t}$ and 2^s into 2^t , with the same property for all $\rho, \tau \in 2^{\leq s}$.

Lemma 3.1.2. *Every node embedding $\theta: 2^{<s} \rightarrow 2^{<t}$ extends to a full embedding. If (b_σ) , (a_η) is a tree of height t in M and θ is a full embedding $2^{\leq s} \rightarrow 2^{\leq t}$, then $(b_{\theta(\sigma)} : \sigma \in 2^{<s})$, $(a_{\theta(\eta)} : \eta \in 2^s)$ is a tree of height s in M .*

Proof. For $\eta \in 2^s$, let $\theta(\eta)$ be any element of 2^t extending $\theta(\eta|(s-1)) \frown \langle \eta(s-1) \rangle$. Suppose that $\rho \frown \langle u \rangle \sqsubseteq \eta$. If $\rho = \eta|(s-1)$, the requirement holds by construction. Otherwise $\rho \frown \langle u \rangle \sqsubseteq \eta|(s-1)$, so $\theta(\rho) \frown \langle u \rangle \sqsubseteq \theta(\eta|(s-1)) \sqsubseteq \theta(\eta)$. The second statement is immediate, since whether $\varphi(a_\eta, b_\sigma)$ holds depends only on the direction u with $\sigma \frown \langle u \rangle \sqsubseteq \eta$, and full embeddings preserve this direction. ☺

In the next proof and in the rest of this chapter, for $u \in \{0, 1\}$ and a finite 0–1 sequence σ , we write $\langle u \rangle \frown \sigma$ for the sequence obtained by putting u in front of σ .

Figure 3.1 shows a node embedding, and the lemma below for $t_0 = t_1 = 2$.

Lemma 3.1.3 (Ramsey lemma for trees). *Let $t_0, t_1 \geq 1$ be given, let $t = t_0 + t_1 - 1$, and suppose that $2^{<t} = C_0 \cup C_1$. Then for some $u \in \{0, 1\}$ there is a node embedding $\theta: 2^{<t_u} \rightarrow 2^{<t}$ with image contained in C_u .*

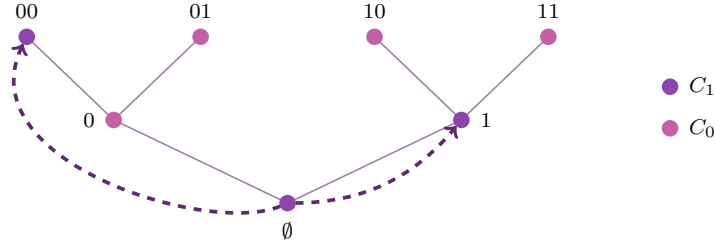


FIGURE 3.1. The nodes $2^{<3}$, growing upward, coloured with two colours C_0 (pink) and C_1 (purple). The dashed arrows show a node embedding $\theta: 2^{<2} \rightarrow 2^{<3}$ into C_1 , with $\theta(\emptyset) = \emptyset$, $\theta(0) = 00$ and $\theta(1) = 1$: the image of the left child lies above the image of the root in the left direction, and the image of the right child in the right direction, but θ need not send children to children. This is the conclusion of Lemma 3.1.3 for $t_0 = t_1 = 2$ and $t = 3$. The colour C_0 admits no such embedding here.

Proof. Induction on t . If $t = 1$, then $t_0 = t_1 = 1$, and $\emptyset \mapsto \emptyset$ works for the u with $\emptyset \in C_u$. Now suppose that $t > 1$, and choose w with $\emptyset \in C_w$. If $t_w = 1$, the map $\emptyset \mapsto \emptyset$ works, so assume that $t_w \geq 2$. For $v \in \{0, 1\}$ and $i \in \{0, 1\}$ let $C_i^v = \{\sigma \in 2^{<t-1} : \langle v \rangle \frown \sigma \in C_i\}$. Then $C_0^v \cup C_1^v$ covers $2^{<t-1}$, and $(t_w - 1) + t_{1-w} - 1 = t - 1$. By induction, for each v either

- (i) there is a node embedding $\chi: 2^{<t_{1-w}} \rightarrow C_{1-w}^v$, in which case $\sigma \mapsto \langle v \rangle \frown \chi(\sigma)$ is a node embedding $2^{<t_{1-w}} \rightarrow 2^{<t}$ with image in C_{1-w} , and we are done; or
- (ii) there is a node embedding $\chi_v: 2^{<t_w-1} \rightarrow C_w^v$.

If (ii) holds for both v , define $\theta: 2^{<t_w} \rightarrow 2^{<t}$ by $\theta(\emptyset) = \emptyset$ and $\theta(\langle v \rangle \frown \sigma) = \langle v \rangle \frown \chi_v(\sigma)$. The image of θ lies in C_w . Suppose that $\rho \frown \langle u \rangle \sqsubseteq \tau$. If $\rho = \emptyset$ then $\tau = \langle u \rangle \frown \sigma$ and $\emptyset \frown \langle u \rangle \sqsubseteq \theta(\tau)$. If $\rho = \langle v \rangle \frown \rho'$ then $\tau = \langle v \rangle \frown \tau'$ with $\rho' \frown \langle u \rangle \sqsubseteq \tau'$, so $\chi_v(\rho') \frown \langle u \rangle \sqsubseteq \chi_v(\tau')$. In both cases $\theta(\rho) \frown \langle u \rangle \sqsubseteq \theta(\tau)$, so θ is a node embedding. $\odot$

Definition 3.1.4. Let $(b_\sigma)_{\sigma \in 2^{<t}}$, $(a_\eta)_{\eta \in 2^t}$ be a tree in M . An *up-ladder of length k* in this tree consists of nodes $\sigma_1, \dots, \sigma_k$ and leaves $\eta_1, \dots, \eta_k$ such that $M \models \varphi(a_{\eta_p}, b_{\sigma_q}) \iff p \leq q$. A *down-ladder of length ℓ* is defined in the same way with “ $p > q$ ” in place of “ $p \leq q$ ”.

The height needed to force a ladder is governed by a familiar recursion. Define $B(k, \ell)$ for $k, \ell \geq 1$ by $B(k, 1) = B(1, \ell) = 1$ and $B(k, \ell) = B(k-1, \ell) + B(k, \ell-1) + 1$ for $k, \ell \geq 2$. By induction on $k + \ell$, using Pascal’s rule $\binom{k+\ell-3}{k-2} + \binom{k+\ell-3}{k-1} = \binom{k+\ell-2}{k-1}$, we have $B(k, \ell) = 2\binom{k+\ell-2}{k-1} - 1$. The recursion is the Erdős–Szekeres recursion for the Ramsey numbers $R(k, \ell)$, and the bound mirrors $R(k, \ell) \leq \binom{k+\ell-2}{k-1}$; the proof below follows the Erdős–Szekeres pattern as well: we fix a leaf η_* and split the nodes according to whether a_{η_*} satisfies the corresponding instance of φ . Lemma 3.1.3 finds a large subtree whose nodes all lie on one side of the split, induction finds a ladder inside that subtree, and the pair formed by η_* and the root of the subtree extends the ladder by one rung.

Proposition 3.1.5. *For all $k, \ell \geq 1$, every tree of height $B(k, \ell)$ for φ in a model of T contains an up-ladder of length k or a down-ladder of length ℓ .*

Proof. Induction on $k + \ell$. In a tree of height 1, the root $\emptyset$ with the leaf $\langle 1 \rangle$ is an up-ladder of length 1, and the root with the leaf $\langle 0 \rangle$ is a down-ladder of length 1. This settles the case $k = 1$ or $\ell = 1$.

Let $k, \ell \geq 2$ be given, let $t = B(k, \ell)$, and let $(b_\sigma), (a_\eta)$ be a tree of height t in M . Fix a leaf $\eta_* \in 2^t$ and put

$$P^+ = \{\sigma \in 2^{<t} : M \models \varphi(a_{\eta_*}, b_\sigma)\}, \quad P^- = 2^{<t} \setminus P^+.$$

Let $t^+ = B(k-1, \ell) + 1$ and $t^- = B(k, \ell-1) + 1$, so that $t^+ + t^- - 1 = t$. By Lemma 3.1.3, one of the two cases below occurs.

Case +: there is a node embedding $\theta: 2^{<t^+} \rightarrow 2^{<t}$ with image in P^+ . Extend θ to a full embedding $\hat{\theta}$ (see Lemma 3.1.2) and set $\psi(\tau) = \hat{\theta}(\langle 0 \rangle \frown \tau)$ for $\tau \in 2^{\leq t^+-1}$. Then ψ is a full embedding of height $B(k-1, \ell)$. Put $\sigma_0 = \theta(\emptyset)$. Since $\emptyset \frown \langle 0 \rangle \sqsubseteq \langle 0 \rangle \frown \tau$, we have $\sigma_0 \frown \langle 0 \rangle \sqsubseteq \psi(\tau)$ for every τ . By Lemma 3.1.2, the embedding ψ induces a tree of height $B(k-1, \ell)$ in M . By induction, the induced tree contains a down-ladder of length ℓ , which is then a down-ladder in the original tree, or an up-ladder $(\sigma'_p, \eta'_p)_{p=1}^{k-1}$. In the latter case, let

$$(\sigma_1, \eta_1) = (\sigma_0, \eta_*), \quad (\sigma_{p+1}, \eta_{p+1}) = (\psi(\sigma'_p), \psi(\eta'_p)) \quad (1 \leq p < k).$$

We check that $\varphi(a_{\eta_p}, b_{\sigma_q}) \iff p \leq q$.

- (a) For $p = 1$, every σ_q is in the image of θ , which lies in P^+ , so $\varphi(a_{\eta_*}, b_{\sigma_q})$ holds, and indeed $1 \leq q$.
- (b) For $p \geq 2$ and $q = 1$, we have $\sigma_0 \frown \langle 0 \rangle \sqsubseteq \eta_p$, so $\neg \varphi(a_{\eta_p}, b_{\sigma_0})$, and indeed $p > 1$.
- (c) For $p, q \geq 2$: this is the up-ladder property of the smaller tree.

This is an up-ladder of length k .

Case -: there is a node embedding $\theta: 2^{<t^-} \rightarrow 2^{<t}$ with image in P^- . Proceed symmetrically, with $\psi(\tau) = \hat{\theta}(\langle 1 \rangle \frown \tau)$, a full embedding of height $B(k, \ell-1)$ whose leaves all extend $\sigma_0 \frown \langle 1 \rangle$. By induction the induced tree contains an up-ladder of length k , and we are done, or a down-ladder $(\sigma'_p, \eta'_p)_{p=1}^{\ell-1}$. Prepend (σ_0, η_*) as before.

- (a) For $p = 1$, the formula $\neg \varphi(a_{\eta_*}, b_{\sigma_q})$ holds for all q , because the nodes lie in P^- , and indeed $1 \not\leq q$.
- (b) For $p \geq 2$ and $q = 1$, the formula $\varphi(a_{\eta_p}, b_{\sigma_0})$ holds, because $\sigma_0 \frown \langle 1 \rangle \sqsubseteq \eta_p$, and indeed $p > 1$.

This is a down-ladder of length ℓ . ☺

Corollary 3.1.6. *If a model M of T contains a tree of height $B(n+1, n+1)$ for φ , then M contains an n -ladder for φ .*

Proof. By Proposition 3.1.5, the model M contains an up-ladder or a down-ladder of length $n+1$. An up-ladder of length $n+1$ is an $(n+1)$ -ladder, and so contains an n -ladder. A down-ladder of length $n+1$ yields an n -ladder by Remark 1.1.5. ☺

3.2. The classical proof of definability, via ranks

In Proposition 3.2.3 below we give the classical argument that, in the absence of tall trees, every φ -type is definable. Together with Proposition 2.3.1, the argument gives another route to (1) $\Rightarrow$ (3) in Theorem 1.1.6, at the cost of positivity: the definitions obtained in this section are not positive Boolean combinations of instances of φ , since they contain quantifiers and negations that do not occur in φ (see Lemma 3.2.2(iv)). The idea is to measure each finite piece of a type by the height of the tallest tree that can be grown above the piece, and to define the type by a piece of minimal height.

Assume that there is n such that no model of T contains a tree of height n , and fix a model M of T . By a φ -conjunction over M we mean a finite conjunction $\theta(x)$ of formulas $\varphi(x, b)^u$ with $b \in M$.

Definition 3.2.1. For a φ -conjunction θ over M and $m < \omega$, we write $R(\theta) \geq m$ if there are $b_\sigma \in M$ ($\sigma \in 2^{<m}$) such that for every $\eta \in 2^m$, we have

$$M \models \exists x \left(\theta(x) \wedge \bigwedge_{k < m} \varphi(x, b_{\eta|k})^{\eta(k)} \right).$$

Thus $R(\theta) \geq 0$ if and only if $M \models \exists x \theta(x)$. When θ is satisfiable in M , we write $R(\theta)$ for the largest m with $R(\theta) \geq m$.

In Lemma 3.2.2(i) below we show that this largest m exists.

Lemma 3.2.2.

- (i) *The inequality $R(\theta) < n$ holds for every θ ; in particular $R(\theta)$ is well defined whenever θ is satisfiable in M .*
- (ii) *If $\theta' = \theta \wedge \chi$, then $R(\theta') \leq R(\theta)$.*
- (iii) *If $b \in M$ and $R(\theta \wedge \varphi(x, b)^u) \geq m$ for both $u \in \{0, 1\}$, then $R(\theta) \geq m + 1$.*
- (iv) *For each θ and m there is an $L(M)$ -formula $\psi_{\theta, m}(y)$ such that for all $b \in M$ we have $R(\theta \wedge \varphi(x, b)) \geq m \iff M \models \psi_{\theta, m}(b)$.*

Proof. (i) If $R(\theta) \geq n$, witnessed by (b_σ) , then choosing a witness $a_\eta \in M$ for each $\eta \in 2^n$ gives a tree of height n in M , which is impossible.

(ii) A witness for $R(\theta') \geq m$ is a witness for $R(\theta) \geq m$.

(iii) Let $(b_\sigma^u)_{\sigma \in 2^{<m}}$ witness $R(\theta \wedge \varphi(x, b)^u) \geq m$. Put $b_\emptyset = b$ and $b_{\langle u \rangle \frown \sigma} = b_\sigma^u$. A branch of length $m + 1$ has the form $\langle u \rangle \frown \eta$ with $\eta \in 2^m$, and the corresponding formula is $\theta \wedge \varphi(x, b)^u \wedge \bigwedge_{k < m} \varphi(x, b_{\eta|k}^u)^{\eta(k)}$, which is satisfiable in M .

(iv) Take

$$\psi_{\theta, m}(y) := \exists (z_\sigma)_{\sigma \in 2^{<m}} \bigwedge_{\eta \in 2^m} \exists x \left(\theta(x) \wedge \varphi(x, y) \wedge \bigwedge_{k < m} \varphi(x, z_{\eta|k})^{\eta(k)} \right),$$

whose parameters are those of θ . ☺

Proposition 3.2.3. *Let M be a model of T . If for some n no model of T contains a tree of height n , then every $p \in S_\varphi(M)$ is definable.*

Proof. Let Φ_p be the set of finite conjunctions of members of p . Each $\theta \in \Phi_p$ is satisfiable in M , because the sentence $\exists x \theta$ is an $L(M)$ -sentence consistent with the elementary diagram of M , hence true in M . By Lemma 3.2.2(i), choose $\theta \in \Phi_p$ with $m = R(\theta)$ minimal. We show that $\psi_{\theta, m}$ is a φ -definition of p , i.e. that for $b \in M$, we have

$$\varphi(x, b) \in p \iff R(\theta \wedge \varphi(x, b)) \geq m.$$

If $\varphi(x, b) \in p$, then $\theta \wedge \varphi(x, b) \in \Phi_p$, so $R(\theta \wedge \varphi(x, b)) \geq m$ by minimality. Conversely, suppose that $\varphi(x, b) \notin p$. Then $\neg \varphi(x, b) \in p$, so $R(\theta \wedge \neg \varphi(x, b)) \geq m$ by minimality. If also $R(\theta \wedge \varphi(x, b)) \geq m$, then Lemma 3.2.2(iii) would give $R(\theta) \geq m + 1$, a contradiction. The equivalence together with Lemma 3.2.2(iv) shows that p is definable. ☺



Remark 3.2.4.

- (a) The arguments are effective. An up- or down-ladder of length $n + 1$ appears in any tree of height $B(n + 1, n + 1) = 2^{\binom{2n}{n}} - 1$. Conversely, a ladder of length $2^{n+1} - 1$ yields a tree of height n .

- (b) Theorem 2.2.1 uses only that p is finitely satisfiable in M (see Lemma 1.1.2). So the theorem applies to any φ -type over a set A that is finitely satisfiable in A . The parameters of that definition come from the finite set F , whereas the rank argument of Proposition 3.2.3 yields a definition whose parameters are those of the formula θ of minimal rank.
- (c) Without finite satisfiability, definitions over the parameter set cannot be expected in general, and the restriction to types over models is essential. Definition 3.2.5 and Example 3.2.6 below make this precise.

Definition 3.2.5. Let N be a model of T , and let $A \subseteq N$ and $p \in S_\varphi(N)$ be given.

- (a) A *complete φ -type over A* is defined as in Definition 1.1.1, with b ranging over A instead of M ; such a type is *finitely satisfiable in A* if every finite subset of the type is realized by an element of A .
- (b) The type p is *definable over A* if p has a φ -definition that is an $L(A)$ -formula.
- (c) The type p is *definable almost over A* if p has a φ -definition of the form $\psi(y, c)$, with $\psi(y, z)$ an L -formula and c a tuple of N , such that only finitely many distinct subsets of N are of the form $\psi(N, c')$ with $c' \in N$ realizing the same complete type over A as c . In other words, the definition of p has only finitely many conjugates over A .

Definable over A implies definable almost over A : take c to enumerate the parameters from A , so that $c' = c$ is the only possibility. The converse fails, as the next example shows with two equivalence classes.

Example 3.2.6. Let T be the theory of an equivalence relation E with exactly two classes, both infinite, let $\varphi(x, y)$ be $E(x, y)$, and let $A = \emptyset$. Let N be a model of T with classes C_1 and C_2 , and let $p \in S_\varphi(N)$ be the φ -type saying that x lies in C_1 , i.e. $E(x, b) \in p$ if and only if $b \in C_1$. For any $c \in C_1$, the formula $E(y, c)$ is a φ -definition of p . All elements of N have the same complete type over $\emptyset$, so the sets $E(N, c')$ are exactly C_1 and C_2 . Hence p is definable almost over $\emptyset$. The type p is not definable over $\emptyset$: since N has an automorphism swapping C_1 and C_2 , every L -formula $\psi(y)$ without parameters defines $\emptyset$ or N , never C_1 . Note that p is not finitely satisfiable in $\emptyset$.

3.3. Historical remarks

- ✂ **Section 3.1.** The Ramsey-type argument that extracts ladders from trees is due to Hodges [53]. Hodges's argument gives a direct combinatorial proof that tall trees force the order property, without going through definability of types. Conant and Terry [45] recently revisited the argument with explicit bounds, and extended the argument from $\{0, 1\}$ -valued formulas to real-valued functions. Their version is the one followed here. Such quantitative bounds are what is needed for applications of stability to finite combinatorics, such as the regularity lemma for stable graphs.
- ✂ **Section 3.2.** The rank $R(\theta)$ is Shelah's local rank $R(\theta, \varphi, 2)$ [36, 57]. The proof that a type of minimal rank is definable is the classical route to definability of types, the one found in most textbook treatments, for instance Pillay [56] and Tent–Ziegler [58], and predates the combinatorial and analytic proofs given elsewhere in these notes. As noted above, the resulting definitions involve quantifiers, whereas those of Section 2.2 do not.

Chapter 4

Stability and notions of independence

Ladders, trees and definitions all describe stability from inside a single formula. Independence describes stability globally: in this chapter we show that the stable theories are exactly those carrying a notion of independence, written $A \downarrow_M B$ and read “ A is independent from B over M ”, that obeys a short list of natural axioms. The most important of these are finite character, symmetry, and the existence and uniqueness of independent extensions. The idea is that, given a type p over a model M and a set B , an independent extension of p to $M \cup B$ adds no information beyond p . Stability is precisely the condition that such an extension always exists and is unique.

Both directions rest on Chapter 2. To obtain independence from stability we use definability of types: the independent extension is the *heir*, the extension of p given by the same definition scheme as p . The converse is a counting argument of the kind used there.

4.1. Conventions

From now on T is a *complete* theory in the language L , and $\kappa = |T| = |L| + \aleph_0$. We work inside a monster model $\mathbb{M}$ of T : a $\bar{\kappa}$ -saturated and strongly $\bar{\kappa}$ -homogeneous model for some very large cardinal $\bar{\kappa}$ (larger than 2^{2^κ} , say). A set is *small* if of cardinality $< \bar{\kappa}$. Letters A, B, C denote small subsets of $\mathbb{M}$; tuples may be infinite, but are always small. We freely identify a set with an enumerating tuple. A *model* is a small elementary substructure $M \preceq \mathbb{M}$. Every model of T of size $< \bar{\kappa}$ embeds elementarily in $\mathbb{M}$, and every type over a small set is realized in $\mathbb{M}$.

We write $\text{tp}(a/A)$ for the complete type of a over A , $S_x(A)$ for the set of complete types over A in the variables x , and $a \equiv_A a'$ for $\text{tp}(a/A) = \text{tp}(a'/A)$. We write MB for $M \cup B$. By strong homogeneity, we have $a \equiv_A a'$ if and only if $\sigma(a) = a'$ for some $\sigma \in \text{Aut}(\mathbb{M}/A)$.

Definition 4.1.1.

- (a) The theory T is *stable* if no L -formula $\varphi(x, y)$ has the order property (see Definition 1.1.4).
- (b) For an infinite cardinal λ , the theory T is λ -*stable* if $|S_x(A)| \leq \lambda$ whenever $|A| \leq \lambda$ and x is a finite tuple of variables.

Lemma 4.1.2. *If T is λ -stable for some $\lambda \geq \kappa$, then T is stable.*

Proof. Suppose that $\varphi(x, y)$ has the order property. By Proposition 2.3.1, the formula φ encodes binary trees of arbitrarily large finite height. By Proposition 2.4.2 there is a model N of T with $|N| \leq \lambda$ and $|S_\varphi(N)| > \lambda$. Since T is complete, we may assume $N \preceq \mathbb{M}$. Every φ -type over N is consistent with the elementary diagram of N , so each φ -type over N extends to a complete type in $S_x(N)$, and distinct φ -types have distinct extensions. Hence $|S_x(N)| > \lambda \geq |N|$, and T is not λ -stable. ☺

4.2. Definitions and heirs

In a stable theory every type over a model has a definition scheme, one definition for each formula (see Lemma 4.2.1). A definition can be evaluated at parameters outside the model, so the scheme extends the type to larger sets of parameters, and the result is the heir (see Definition 4.2.2). The section closes with the fact that will make independence symmetric: definitions over a model commute (see Lemma 4.2.4). In Subsection 4.2.1 we show that this commutation is the exchange of iterated ultrafilter limits that defines stability in Section 1.2.

Lemma 4.2.1. *Assume that T is stable, and let $p \in S_x(M)$ be given, where x may be infinite. For every $L(M)$ -formula $\varphi(x', y)$, with x' a finite subtuple of x and y finite, there is an $L(M)$ -formula $d_p\varphi(y)$ such that for all $b \in M$, we have*

$$\varphi(x', b) \in p \iff M \models d_p\varphi(b).$$

Any two formulas with this property are equivalent in $\mathbb{M}$.

Proof. Write $\varphi(x', y) = \varphi_0(x', y, m)$ with $\varphi_0 \in L$ and $m \in M$. The set of formulas $\varphi_0(x', c)^u$ ($c \in M$) that belong to p is a complete φ_0 -type over M in the sense of Definition 1.1.1. Since φ_0 does not have the order property, Theorem 1.1.6 shows that this φ_0 -type has a definition $\psi(z)$ over M , which is a positive Boolean combination of instances of φ_0 . Here z corresponds to the parameter variables y followed by those replaced by m . Then $d_p\varphi(y) := \psi(y, m)$ works. If $\psi_1(y)$ and $\psi_2(y)$ both work, then $M \models \forall y (\psi_1 \leftrightarrow \psi_2)$. This is an $L(M)$ -sentence, so this sentence also holds in $\mathbb{M}$. 

The family $(d_p\varphi)_\varphi$ given by Lemma 4.2.1, determined up to equivalence in $\mathbb{M}$, is called the *definition scheme* of p .

Definition 4.2.2. Assume that T is stable. Let $p \in S_x(M)$ be given, and let B be small. Define

$$p|B := \{\varphi(x', b) : \varphi(x', y) \in L(M), b \in B, \mathbb{M} \models d_p\varphi(b)\}.$$

The set $p|B$ is called the *heir* of p over MB .

Every $L(MB)$ -formula has the form $\varphi(x', b)$ with $\varphi \in L(M)$ and $b \in B$, so $p|B$ is a set of $L(MB)$ -formulas.

Lemma 4.2.3. *Assume that T is stable, let $p \in S_x(M)$ be given, and let B and D be small sets. Then the following hold.*

- (a) *The set $p|B$ belongs to $S_x(MB)$ and extends p .*
- (b) *If $B \subseteq D$, then $p|B$ is the restriction of $p|D$ to MB .*
- (c) *If N is a model with $M \subseteq N$, then $p|N$ has the same definition scheme as p , and $(p|N)|B = p|NB$.*

Proof. (a) *Completeness.* Let $\varphi \in L(M)$ be given. Since p is complete, for every $c \in M$ exactly one of $\varphi(x', c)$, $\neg\varphi(x', c)$ lies in p . Hence $M \models \forall y (d_p(\neg\varphi)(y) \leftrightarrow \neg d_p\varphi(y))$, and the same holds in $\mathbb{M}$. So for $b \in B$ exactly one of $\varphi(x', b)$, $\neg\varphi(x', b)$ belongs to $p|B$.

Consistency. Let $\varphi_i(x_i, b_i) \in p|B$ be given for $i < n$, and put $\chi(\bar{x}; y_0, \dots, y_{n-1}) = \bigwedge_{i < n} \varphi_i(x_i, y_i)$, where $\bar{x}$ lists the variables in the x_i . For $\bar{c} \in M$, the formula $\chi(\bar{x}, \bar{c})$ is in p if and only if each $\varphi_i(x_i, c_i)$ is, and in that case $\chi(\bar{x}, \bar{c})$ is satisfiable in M . Therefore

$$M \models \forall \bar{y} \left(\bigwedge_{i < n} d_p\varphi_i(y_i) \rightarrow d_p\chi(\bar{y}) \right) \quad \text{and} \quad M \models \forall \bar{y} (d_p\chi(\bar{y}) \rightarrow \exists \bar{x} \chi(\bar{x}, \bar{y})).$$

Both sentences transfer to $\mathbb{M}$. Applied to $\bar{b} = (b_0, \dots, b_{n-1})$, they give $\mathbb{M} \models \exists \bar{x} \bigwedge_i \varphi_i(x_i, b_i)$. By compactness, the set $p|B$ is consistent.

Extension. For $c \in M$ we have $\varphi(x', c) \in p \iff M \models d_p\varphi(c)$, which holds if and only if $\varphi(x', c) \in p|B$.

(b) This is immediate from the definition.

(c) By (a), the type $p|N$ extends p . For $n \in N$ we have $\varphi(x', n) \in p|N$ if and only if $M \models d_p\varphi(n)$, so $d_p\varphi$ is a definition of $p|N$ for $\varphi \in L(M)$. For an $L(N)$ -formula $\varphi(x', y) = \varphi_0(x', y, n)$ with $\varphi_0 \in L(M)$, the formula $d_p\varphi_0(y, n)$ defines $p|N$. By the uniqueness clause of Lemma 4.2.1, the definition scheme of $p|N$ is thus given, up to equivalence, by that of p . Now for $b \in B$ and $n \in N$, we have

$$\varphi_0(x', b, n) \in (p|N)|B \iff M \models d_p\varphi_0(b, n) \iff \varphi_0(x', b, n) \in p|NB. \quad \text{☺}$$

The heart of the matter is the following lemma, which says that definitions of types over a model commute; this lemma is the only place in this chapter where the absence of the order property is used directly. To state the lemma, let $p \in S_x(M)$ and $q \in S_y(M)$ be given, with x and y finite, and let $\varphi(x, y) \in L(M)$ be given. As before, we write $d_p\varphi(y)$ for the definition of p for $\varphi(x, y)$, and we write $d_q\varphi(x)$ for the definition of q for φ regarded as a formula in the variables y , with parameter variables x . In analytic terms, the two sides of Harrington's lemma say that the two iterated limits of φ , taken in the two possible orders, equal 1, and the lemma says that these iterated limits agree; we make this precise in Subsection 4.2.1, right after the proof.

Lemma 4.2.4 (Harrington's lemma). *Assume that T is stable, and let $p \in S_x(M)$, $q \in S_y(M)$ and $\varphi(x, y) \in L(M)$ be given, where x and y are finite. Then*

$$d_p\varphi(y) \in q \iff d_q\varphi(x) \in p.$$

Proof. The idea is to realize p and q alternately, so that each pair (a_i, b_j) consults one definition below the diagonal and the other on and above the diagonal; if the two definitions disagreed, the result would be a ladder. Choose recursively $a_i, b_i \in M$ ($i < \omega$) such that a_i realizes $p|b_j : j < i$ and b_i realizes $q|a_j : j \leq i$. This is possible by Lemma 4.2.3(a) and saturation. In particular $a_i \models p$ and $b_i \models q$.

If $j < i$, then since $a_i \models p|b_j$ and $b_j \models q$, we have

$$M \models \varphi(a_i, b_j) \iff M \models d_p\varphi(b_j) \iff d_p\varphi(y) \in q.$$

If $i \leq j$, then since $b_j \models q|a_i$ and $a_i \models p$, we have

$$M \models \varphi(a_i, b_j) \iff M \models d_q\varphi(a_i) \iff d_q\varphi(x) \in p.$$

Suppose the two right-hand sides differ, and write $\varphi(x, y) = \varphi_0(x, y, m)$ with $\varphi_0 \in L$ and $m \in M$. Put $b'_j = b_j m$. Then either $M \models \varphi_0(a_i, b'_j) \iff i \leq j$ for all i, j , or $M \models \varphi_0(a_i, b'_j) \iff i > j$ for all i, j . In the first case $\varphi_0(x; y, z)$ has the order property directly, and in the second by Remark 1.1.5. Either way this contradicts stability. ☺

Figure 4.1 shows the configuration built in the proof.

4.2.1. Harrington's lemma as limit exchange

Harrington's lemma is the exchange of iterated ultrafilter limits of Section 1.2, read in the language of types. We now make this precise, and in doing so obtain a second proof of the lemma that does not build a ladder by hand.

Let x and y be finite tuples of variables. Every type $p \in S_x(M)$ is finitely satisfiable in M : if $\psi(x) \in p$, then we have $M \models \exists x \psi(x)$, and since ψ is an $L(M)$ -formula and $M \preceq M$, some $a \in M^{|x|}$ satisfies ψ . Hence the sets $\psi(M) = \{a \in M^{|x|} : M \models \psi(a)\}$, for $\psi \in p$, have

The computation also explains what the two orders of realization mean. For a small set B , the *coheir* of $\mathcal{U}$ over MB is the type

$$q_{\mathcal{U}}|B = \{\varphi(x, b) : \varphi(x, y) \in L(M), b \in B, \{a \in M^{|x|} : \mathbb{M} \models \varphi(a, b)\} \in \mathcal{U}\},$$

a complete type over MB that is finitely satisfiable in M ; every type over MB that is finitely satisfiable in M arises in this way. If b realizes $q_{\mathcal{V}}|\emptyset$ and a realizes $q_{\mathcal{U}}|\{b\}$, then unwinding the definitions gives

$$\mathbb{M} \models \varphi(a, b) \iff \lim_{a', \mathcal{U}} \lim_{b', \mathcal{V}} \varphi(a', b') = 1,$$

and realizing the two coheirs in the other order replaces this iterated limit by the one in the other order. So the two orders of realizing coheirs give the same type exactly when the iterated limits commute. In a stable theory, moreover, the coheir $q_{\mathcal{U}}|B$ is the heir of $q_{\mathcal{U}}|\emptyset$, because both are nonforking extensions and nonforking extensions of types over models are unique (see Remark 4.4.2 and [56, 58]). Read in this way, the symmetry of independence proved in Section 4.3 is the commutation of ultrafilter limits: the logical counterpart of exchanging the order of two limits in analysis.

4.3. Independence relations

The axioms below record how the heir relation behaves. When T is stable the heir relation satisfies all of them, and conversely a relation satisfying just two of them, uniqueness and local character, forces T to be stable (see Theorem 4.3.3).

Definition 4.3.1. An *independence relation (over models)* is a relation $A \downarrow_M B$ between small sets A, B and models M satisfying the following axioms, for all small A, A', B, B' and all models M, N .

- (INV) *Invariance.* If $A \downarrow_M B$ and $\sigma \in \text{Aut}(\mathbb{M})$, then $\sigma(A) \downarrow_{\sigma(M)} \sigma(B)$.
- (MON) *Monotonicity.* If $A \downarrow_M B$, $A' \subseteq A$ and $B' \subseteq B$, then $A' \downarrow_M B'$.
- (BMON) *Base monotonicity.* If $A \downarrow_M B$ and N is a model with $M \subseteq N \subseteq B$, then $A \downarrow_N B$.
- (TR) *Transitivity.* If $M \subseteq N$, $A \downarrow_M N$ and $A \downarrow_N B$, then $A \downarrow_M NB$.
- (FC) *Finite character.* The relation $A \downarrow_M B$ holds if and only if $a \downarrow_M b$ for all finite tuples a from A and b from B .
- (SYM) *Symmetry.* The relation $A \downarrow_M B$ holds if and only if $B \downarrow_M A$.
- (EX) *Existence (extension).* For all A, M, B there is $A' \equiv_M A$ with $A' \downarrow_M B$.
- (UN) *Uniqueness (stationarity).* If $A \equiv_M A'$, $A \downarrow_M B$ and $A' \downarrow_M B$, then $A \equiv_{MB} A'$.
- (LOC) *Local character.* For every finite tuple a and every model N there is a model $M \subseteq N$ with $|M| \leq \kappa$ and $a \downarrow_M N$.

In terms of types, (EX) and (UN) say that a type $p \in S_x(M)$ has exactly one *independent extension* to MB , namely the type over MB of any $A \models p$ with $A \downarrow_M B$. All the other axioms are statements about which extensions of p count as independent.

Definition 4.3.2. Assume that T is stable. For small A, B and a model M we define

$$A \downarrow_M B \iff \text{tp}(A/MB) = \text{tp}(A/M)|B.$$

In words, the type $\text{tp}(A/MB)$ is the *heir* of $\text{tp}(A/M)$, the extension defined by the same definition scheme.

For finite tuples a, b with $p = \text{tp}(a/M)$, every $L(Mb)$ -formula in x has the form $\varphi(x, b)$ with $\varphi(x, y) \in L(M)$. So

$$(4.1) \quad a \downarrow_M b \iff \text{for all } \varphi(x, y) \in L(M) : \quad \mathbb{M} \models \varphi(a, b) \leftrightarrow d_p \varphi(b).$$

Theorem 4.3.3. *For a complete theory T the following are equivalent.*

- (1) *The theory T is stable.*
- (2) *There is an independence relation satisfying all the axioms of Definition 4.3.1.*
- (3) *There is a relation $A \downarrow_M B$ satisfying (UN) and (LOC).*

Moreover, if T is stable, the relation of Definition 4.3.2 satisfies all the axioms. This relation also satisfies

- (AR) Anti-reflexivity. *If $a \downarrow_M a$ for a finite tuple a , then $a \subseteq M$.*

4.3.1. Proof of (1) $\Rightarrow$ (2)

Assume that T is stable, and let $\downarrow$ be the relation of Definition 4.3.2. Throughout, we write $p = \text{tp}(A/M)$. Most of the axioms follow by unwinding the definitions with the help of Lemma 4.2.3; symmetry is the one place where Harrington's lemma is needed.

(INV). Let $\sigma \in \text{Aut}(\mathbb{M})$ be given. Applying σ to the parameters of the definition scheme of p gives a definition scheme for $\sigma(p) = \text{tp}(\sigma A/\sigma M)$. Hence $\sigma(p)|\sigma(B) = \sigma(p|B)$. If $\text{tp}(A/MB) = p|B$, then $\text{tp}(\sigma A/\sigma M \sigma B) = \sigma(p|B) = \sigma(p)|\sigma B$.

(MON). The definition scheme of $\text{tp}(A'/M)$ is the part of that of p that involves only variables for A' . So $\text{tp}(A'/M)|B'$ is the restriction of $p|B$ to these variables and to MB' (see Lemma 4.2.3(b)), which is $\text{tp}(A'/MB')$.

(FC). The condition $\text{tp}(A/MB) = p|B$ says that for every $L(M)$ -formula $\varphi(x', y)$ and every $b \in B$ we have $\mathbb{M} \models \varphi(a', b) \leftrightarrow d_p \varphi(b)$, where a' is the finite subtuple of A corresponding to x' . Each instance involves only finite $a' \subseteq A$ and $b \subseteq B$, and is exactly an instance of $a' \downarrow_M b$.

(BMON). Let N be a model with $M \subseteq N \subseteq B$, and suppose that $\text{tp}(A/MB) = p|B$. Then $\text{tp}(A/N) = p|N$ by Lemma 4.2.3(b). By Lemma 4.2.3(c), we have $\text{tp}(A/N)|B = (p|N)|B = p|NB = p|B = \text{tp}(A/NB)$, so $A \downarrow_N B$.

(TR). Let N be a model with $M \subseteq N$, and suppose that $\text{tp}(A/N) = p|N$ and $\text{tp}(A/NB) = \text{tp}(A/N)|B$. By Lemma 4.2.3(c), we have $\text{tp}(A/NB) = (p|N)|B = p|NB$, which means that $A \downarrow_M NB$.

(SYM). By (FC) we need only treat finite tuples a, b . Let $p = \text{tp}(a/M)$ and $q = \text{tp}(b/M)$, and assume that $a \downarrow_M b$. For $\varphi(x, y) \in L(M)$, using (4.1) and Lemma 4.2.4, we have

$$\mathbb{M} \models \varphi(a, b) \iff \mathbb{M} \models d_p \varphi(b) \iff d_p \varphi(y) \in q \iff d_q \varphi(x) \in p \iff \mathbb{M} \models d_q \varphi(a).$$

By (4.1) with the roles of a and b exchanged, this says $b \downarrow_M a$.

(EX). By Lemma 4.2.3(a) and saturation, the type $p|B$ has a realization A' . Then $A' \equiv_M A$ and $A' \downarrow_M B$.

(UN). If $A \equiv_M A'$, then both have type p over M . If both are independent from B over M , then $\text{tp}(A/MB) = p|B = \text{tp}(A'/MB)$.

(LOC). Let a be finite, let N be a model, and let $p = \text{tp}(a/N)$. For each L -formula $\varphi(x, y)$ fix a definition $d_p \varphi(y) \in L(N)$, and let $D \subseteq N$ be the set of all parameters used, so $|D| \leq \kappa$. By downward Löwenheim–Skolem there is a model $M \preceq N$ with $D \subseteq M$ and $|M| \leq \kappa$. Let $q = \text{tp}(a/M)$. Each $d_p \varphi$ is an $L(M)$ -formula, and on M it agrees with the definition of p , so $d_p \varphi$ defines q for φ . These formulas therefore form a definition scheme for q (for $L(M)$ -formulas $\varphi_0(x, y, m)$ use $d_p \varphi_0(y, m)$), and so the heir $q|N$ is defined by the same formulas as p . Since $q|N$ and p are both complete types over N , we have $q|N = p = \text{tp}(a/N)$, which means that $a \downarrow_M N$.

(AR). Suppose that $a \downarrow_M a$ with $p = \text{tp}(a/M)$, and suppose some coordinate a_0 of a is not in M . Let $\varphi(x, y)$ be the formula $x_0 = y_0$, where x_0 and y_0 are the corresponding coordinates of x and y . For $c \in M$ we have $\varphi(x, c) \notin p$, because $a_0 \neq c_0$. So $M \models \neg \exists y d_p \varphi(y)$, and the same holds in $\mathbb{M}$. By (4.1), we have $\mathbb{M} \models \varphi(a, a) \leftrightarrow d_p \varphi(a)$. But $\varphi(a, a)$ holds and $d_p \varphi(a)$ fails, a contradiction.

4.3.2. Proof of (2) $\Rightarrow$ (3) $\Rightarrow$ (1)

The implication (2) $\Rightarrow$ (3) is immediate, because a relation satisfying all the axioms satisfies (UN) and (LOC) in particular. For (3) $\Rightarrow$ (1), the idea is that (LOC) lets us describe each type over a model N by a type over a small submodel, and (UN) guarantees that this description loses no information; counting the descriptions then bounds the number of types. Let $\downarrow$ satisfy (UN) and (LOC), and put $\lambda = 2^\kappa$, so that $\lambda^\kappa = \lambda$. By Lemma 4.1.2, it suffices to show that the theory T is λ -stable.

Let A be a set with $|A| \leq \lambda$, and let x be a finite tuple of variables. By downward Löwenheim–Skolem there is a model $N \supseteq A$ with $|N| \leq \lambda$, and restriction maps $S_x(N)$ onto $S_x(A)$. So we need only show $|S_x(N)| \leq \lambda$.

For each $p \in S_x(N)$ choose a realization a_p . By (LOC) choose a model $M_p \subseteq N$ with $|M_p| \leq \kappa$ and $a_p \downarrow_{M_p} N$. Consider the map

$$p \mapsto (M_p, \text{tp}(a_p/M_p)).$$

This map is injective. Indeed, suppose that $M_p = M_{p'} =: M$ and $a_p \equiv_M a_{p'}$. Since $a_p \downarrow_M N$ and $a_{p'} \downarrow_M N$, (UN) gives $a_p \equiv_N a_{p'}$, i.e. $p = p'$.

The number of subsets of N of size $\leq \kappa$ is at most $\lambda^\kappa = \lambda$. For each such M we have $|S_x(M)| \leq 2^\kappa = \lambda$. Hence $|S_x(N)| \leq \lambda \cdot \lambda = \lambda$. $\odot$

4.4. Remarks

 **Remark 4.4.1.** The proof of (3) $\Rightarrow$ (1) uses only uniqueness and local character. Local character may be weakened to $|M| < \kappa_0$ for any fixed cardinal κ_0 , by taking $\lambda = 2^{\kappa_0 + \kappa}$. So stability is equivalent to the existence of a relation for which types over models have unique independent extensions and every type is independent over a small submodel. All the other axioms hold automatically once T is stable.

 **Remark 4.4.2.** In a stable theory the relation of Definition 4.3.2 is *nonforking independence* restricted to models as bases. For $M \subseteq B$, the following are equivalent: the type $\text{tp}(a/B)$ does not fork over M ; this type is the heir of $\text{tp}(a/M)$; this type is definable over M ; see Pillay [56] or Tent–Ziegler [58]. Nonforking makes sense over arbitrary base sets C . Over such sets, nonforking satisfies all the axioms above except uniqueness, which holds only over algebraically closed sets in T^{eq} (Harnik–Harrington [54]). Example 3.2.6 shows the failure over $\emptyset$. In that theory there is only one 1-type over $\emptyset$, and it has at least two nonforking extensions to N : the type of a new element of C_1 and the type of a new element of C_2 . Their φ -parts are defined by the conjugate formulas $E(y, c)$ with $c \in C_1$ and with $c \in C_2$, so both types are definable almost over $\emptyset$, and neither is definable over $\emptyset$. Kim and Pillay [55] proved that in a stable (indeed in a simple) theory, nonforking is the *only* relation on arbitrary small sets satisfying invariance, monotonicity, transitivity, finite character, symmetry, extension, local character and the independence theorem. Developing forking over arbitrary sets requires considerably more machinery, which is why we have taken models as base sets.

4.5. Historical remarks

✂ **Abstract independence.** The axiomatic study of independence is older than model theory. Steinitz, in his foundational paper on the algebraic theory of fields [59], introduced transcendence bases and showed that any two transcendence bases have the same cardinality. The argument is an exchange argument of the kind now called the Steinitz exchange lemma, the same one that gives the dimension of a vector space. Van der Waerden, in *Moderne Algebra* [60], observed that linear dependence and algebraic dependence obey the same few rules, and derived the existence of bases and the invariance of their cardinality from an abstract notion of dependence whose central axiom is exchange. Whitney [61] went a step further and introduced *matroids*, structures that isolate the abstract properties of linear dependence, and gave several equivalent axiomatizations, in terms of independent sets, bases, circuits and rank. Nakasawa developed the same theory independently in the mid-1930s, and Birkhoff [62] and Mac Lane [63] related matroid theory to lattices and to projective geometry.

After a period in which matroids were studied mainly by Rado and Tutte, Rota made them central to combinatorics. In his monograph with Crapo [64] they become *combinatorial geometries*, and a closure operator with the exchange property is called a *pregeometry*, the term still used in model theory. Rota's view was that dependence is a geometric notion with invariants of its own, such as the Möbius function and the characteristic polynomial of the associated lattice.

Model theory met these ideas in the study of uncountably categorical theories. Baldwin and Lachlan [65] showed that in a strongly minimal set algebraic closure has the exchange property, so algebraic closure is a pregeometry and every model has a well-defined dimension. They used this to give a new proof of Morley's categoricity theorem [35]. Shelah's forking [57] carries independence far beyond pregeometries: nonforking independence is not given by a closure operator, yet in a stable theory nonforking independence has the symmetry, existence and uniqueness properties on which arguments about dimension depend. The later work of Zilber and Hrushovski on the geometry of forking turned the study of forking into a subject of its own, geometric stability theory [56]. The point of view of this chapter, which characterizes independence axiomatically, as van der Waerden and Whitney did for linear dependence, is that of Harnik and Harrington and of Kim and Pillay, discussed below.

✂ **Section 4.1.** Working inside a single large saturated and strongly homogeneous model goes back to Shelah [57], who wrote $\mathfrak{C}$ for this model. The name *monster model* became standard later. The notion of ω -stability comes from Morley's categoricity theorem [35]. The study of λ -stability for arbitrary λ , including the fact that stability is equivalent to λ -stability for all λ with $\lambda^{|T|} = \lambda$, is due to Shelah [36, 57].

✂ **Section 4.2.** Heirs and coheirs (*héritiers* and *cohéritiers*) were introduced by Lascar and Poizat [33]. They gave an alternative presentation of Shelah's forking in stable theories, based only on definability of types and finite satisfiability. Lemma 4.2.4, which says that the definitions of two types over a model commute, is due to Harrington. The lemma appears in Harnik–Harrington [54], and is the key step in proving symmetry of independence in stable theories.

✂ **Section 4.3.** Nonforking independence was introduced by Shelah [57]. Lascar and Poizat [33] reformulated nonforking independence in terms of heirs and definability. The characterization of stability through abstract independence notions is due to Harnik and Harrington [54]. Kim and Pillay [55] later extended this characterization from stable to simple theories. They proved that the simple theories are exactly those carrying a relation satisfying the axioms listed above, with the independence theorem in place of uniqueness, and that such a relation

is then nonforking. Theorem 4.3.3 is the stable case over models, with uniqueness (stationarity over models) as the key axiom. The anchor symbol $\perp$ for independence was introduced by Makkai in a lecture in Jerusalem, during the Model Theory Year at the Institute for Advanced Studies (1980–81), and was adopted at once. The symbol appears in print in Makkai’s survey [48].

Chapter 5

From stability to NIP

In a stable theory no formula can encode arbitrarily long ladders. Many of the best understood structures in mathematics fail this test for a simple reason: they carry a definable linear order. Dense linear orders, the ordered field of real numbers and, more generally, o-minimal structures are all unstable, and yet their definable sets are remarkably well behaved. To account for this kind of tameness we introduce the *independence property*, a configuration much wilder than a ladder, and study the formulas and theories that avoid the independence property, called *NIP*. Every stable formula is NIP, and so are the orders just mentioned, so NIP generalizes both stability and the tameness of order. In Section 5.1 we keep the conventions of Section 1.1, and in Section 5.2 those of Section 1.2.

5.1. The independence property

A ladder (see Definition 1.1.4) is a special pattern: the element a_i satisfies $\varphi(x, b_j)$ exactly for $j \geq i$. The independence property asks instead that *every* pattern be realized.

Definition 5.1.1. Let M be a model of T , and let $n \geq 1$ be given.

- (a) Tuples $b_1, \dots, b_n$ of M are *shattered by φ in M* if for every $S \subseteq \{1, \dots, n\}$ there is a tuple a_S of M such that, for all $1 \leq i \leq n$, we have

$$M \models \varphi(a_S, b_i) \iff i \in S.$$

- (b) The formula φ has the *independence property* if for every n some model of T contains n tuples shattered by φ . Otherwise φ is *NIP* (“not the independence property”); NIP formulas are also called *dependent*.
- (c) The *VC dimension* of φ is the largest n such that some model of T contains n tuples shattered by φ , and ∞ if there is no largest such n .

A theory is *NIP* if every formula is NIP.

Thus φ is NIP if and only if the VC dimension of φ is finite. The terminology comes from learning theory, where shattering is a property of a family of sets. For $a \in M$ we write $\varphi(a, M) = \{b \in M : M \models \varphi(a, b)\}$. If $b_1, \dots, b_n$ are shattered by φ in M , then every subset of $\{b_1, \dots, b_n\}$ is cut out by one of the sets $\varphi(a, M)$, so that the family $\{\varphi(a, M) : a \in M\}$ shatters $\{b_1, \dots, b_n\}$ in the sense of learning theory. As for ladders, an $(n+1)$ -shattered family contains an n -shattered one, so by compactness φ has the independence property if and only if some model of T contains an infinite sequence $(b_i)_{i < \omega}$ such that for every $S \subseteq \omega$ there is a_S with $\varphi(a_S, b_i)$ if and only if $i \in S$.

Proposition 5.1.2. *If φ does not have the order property, then φ is NIP.*

Proof. Suppose that $b_1, \dots, b_n$ are shattered by φ in M , witnessed by the tuples a_S . For $1 \leq i \leq n$ put $S_i = \{j : i \leq j \leq n\}$ and $a_i = a_{S_i}$. Then $M \models \varphi(a_i, b_j)$ if and only if $j \in S_i$, that is, if and only if $i \leq j$. So the suffixes S_i cut out a staircase pattern, and $a_1, \dots, a_n$ and $b_1, \dots, b_n$ form an n -ladder. If φ has the independence property, then φ also has the order property. ☺

Figure 5.1 shows the proof for $n = 3$.

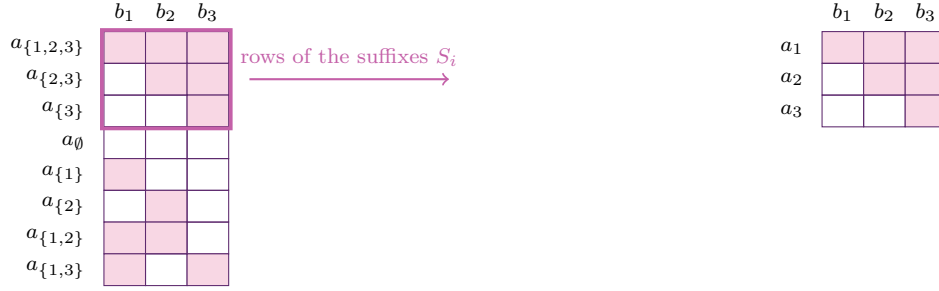


FIGURE 5.1. The proof of Proposition 5.1.2 for $n = 3$. Left: shattered tuples b_1, b_2, b_3 , with one row a_S for each $S \subseteq \{1, 2, 3\}$ and a pink cell where $\varphi(a_S, b_j)$ holds. Right: the three rows indexed by the suffixes $S_1 = \{1, 2, 3\}$, $S_2 = \{2, 3\}$ and $S_3 = \{3\}$ (framed) form the staircase of a 3-ladder (compare Figure 1.1). A ladder uses only n of the 2^n rows, which is why the order property does not give the independence property (Example 5.1.3).

The converse fails, and the first example of the chapter shows why NIP is the right notion for ordered structures.

Example 5.1.3. Let T be the theory of $(\mathbb{Q}, <)$ and let $\varphi(x, y)$ be $x < y$. Then $a_i = i$ and $b_j = j + \frac{1}{2}$ satisfy $\varphi(a_i, b_j)$ if and only if $i \leq j$, so φ has the order property and is unstable. But φ is NIP, with VC dimension 1. Indeed, let b_1, b_2 be elements of a model, say $b_1 \leq b_2$. Shattering requires some a with $a < b_1$ and $a \not< b_2$, that is, some a with $b_2 \leq a < b_1$, which is impossible. So no two elements are shattered. A single element b is shattered, by any $a < b$ and any $a \geq b$.

Example 5.1.4. Let T be the theory of the random graph, whose models are the graphs in which, for all disjoint finite sets U, V of vertices, some vertex outside $U \cup V$ is adjacent to every vertex of U and to no vertex of V . Let $\varphi(x, y)$ be the edge relation $E(x, y)$. Any n distinct vertices $b_1, \dots, b_n$ are shattered: for $S \subseteq \{1, \dots, n\}$ apply the extension property to $U = \{b_i : i \in S\}$ and $V = \{b_i : i \notin S\}$. So E has the independence property.

The independence property of $\varphi(x, y)$ refers to shattering tuples on the y -side. In Proposition 5.1.5 below we show that the choice of side does not matter, at the price of an exponential loss in the number of shattered tuples. We write $\varphi^{\text{opp}}(y, x)$ for $\varphi(x, y)$ with the roles of the variables exchanged.

Proposition 5.1.5. *The formula φ is NIP if and only if φ^{opp} is NIP. More precisely, if some model contains 2^n tuples shattered by φ , then that model contains n tuples shattered by φ^{opp} .*

Proof. Index the 2^n tuples as b_T with $T \subseteq \{1, \dots, n\}$, and let the tuples a_S with $S \subseteq \mathcal{P}(\{1, \dots, n\})$ witness that they are shattered by φ . For $1 \leq i \leq n$ put $S_i = \{T : i \in T\}$ and $a_i = a_{S_i}$. Then $M \models \varphi(a_i, b_T)$ if and only if $T \in S_i$, that is, if and only if $i \in T$. So for every $T \subseteq \{1, \dots, n\}$ the tuple b_T satisfies $\varphi^{\text{opp}}(b_T, a_i)$ exactly for $i \in T$, and $a_1, \dots, a_n$ are shattered by φ^{opp} . Applying this in both directions gives the equivalence. ☺



Remark 5.1.6. A formula $\psi(x, y)$ has the *strict order property* if for every n some model M of T contains tuples $b_1, \dots, b_n$ with $\psi(M, b_1) \subsetneq \psi(M, b_2) \subsetneq \dots \subsetneq \psi(M, b_n)$, where $\psi(M, b) = \{a \in M : M \models \psi(a, b)\}$. The formula $x < y$ of Example 5.1.3 is the basic case. Shelah proved that a theory is unstable if and only if some formula has the independence property or some formula has the strict order property [57, Chapter II]. So an unstable theory either encodes arbitrary subsets, as the random graph does, or encodes an infinite order; NIP theories are those in which only the second reason can occur.

5.2. The real-valued case

As in Section 1.2, formulas give way to bounded functions on arbitrary sets. On arbitrary sets compactness is not available, so shattering is required only for finite patterns, with a uniform gap between the values on the two sides.

Definition 5.2.1. Let A, B be sets and $\varphi: A \times B \rightarrow \mathbb{R}$ a bounded function. A sequence $(b_n)_{n < \omega}$ in B is *independent for φ* if there are reals $r < s$ such that for all disjoint finite sets $P, Q \subseteq \omega$ there is $a \in A$ with

$$\varphi(a, b_n) \geq s \text{ for } n \in P \quad \text{and} \quad \varphi(a, b_n) \leq r \text{ for } n \in Q.$$

The function φ has the *independence property* if some sequence in B is independent for φ , and φ is *NIP* otherwise.

For a $\{0, 1\}$ -valued function, any $0 < r < s < 1$ may be used, and a sequence is independent exactly when each finite part of the sequence is shattered in the sense of Definition 5.1.1. By compactness, a formula $\varphi(x, y)$ is NIP in the sense of Definition 5.1.1 if and only if the corresponding function on $M^{|x|} \times M^{|y|}$ is NIP for every model M of T .

As in the $\{0, 1\}$ -valued case (see Proposition 5.1.2), stability implies NIP. The proof there turned a shattered set into a ladder; here the ladder appears among the types, and the separately continuous extension of Proposition 1.2.8 rules it out.

Proposition 5.2.2. *Every stable function $\varphi: A \times B \rightarrow [0, 1]$ is NIP.*

Proof. Let $\hat{\varphi}: \text{LS}(\varphi) \times \text{RS}(\varphi) \rightarrow [0, 1]$ be the separately continuous function of Proposition 1.2.8. Taking a constant family in (1.2) gives $\hat{\varphi}(p, \text{rtp}_\varphi(b)) = p(b)$ for all $p \in \text{LS}(\varphi)$ and $b \in B$.

Suppose, towards a contradiction, that (b_n) is independent for φ , with reals $r < s$. For $m < N$ choose $a_{m,N} \in A$ with

$$\varphi(a_{m,N}, b_n) \geq s \text{ for } m \leq n < N \quad \text{and} \quad \varphi(a_{m,N}, b_n) \leq r \text{ for } n < m,$$

and let $a_{m,N}$ be arbitrary for $N \leq m$. Fix a nonprincipal ultrafilter $\mathcal{U}$ on ω and put

$$p_m = \lim_{N, \mathcal{U}} \text{ltp}_\varphi(a_{m,N}) \in \text{LS}(\varphi).$$

For fixed n the displayed inequalities hold for all $N > \max(m, n)$, so

$$(5.1) \quad p_m(b_n) \geq s \text{ if } n \geq m, \quad p_m(b_n) \leq r \text{ if } n < m.$$

Now put $q = \lim_{n, \mathcal{U}} \text{rtp}_\varphi(b_n) \in \text{RS}(\varphi)$ and $p = \lim_{m, \mathcal{U}} p_m \in \text{LS}(\varphi)$. We compute $\hat{\varphi}(p, q)$ in two ways.

First, by continuity of $\hat{\varphi}(p_m, \cdot)$ and (5.1), we have $\hat{\varphi}(p_m, q) = \lim_{n, \mathcal{U}} p_m(b_n) \geq s$ for every m , since $p_m(b_n) \geq s$ for all $n \geq m$. By continuity of $\hat{\varphi}(\cdot, q)$, we obtain $\hat{\varphi}(p, q) = \lim_{m, \mathcal{U}} \hat{\varphi}(p_m, q) \geq s$.

Second, since limits in $\text{LS}(\varphi)$ are computed coordinatewise, we have $p(b_n) = \lim_{m, \mathcal{U}} p_m(b_n) \leq r$ for every n , because $p_m(b_n) \leq r$ for all $m > n$. By continuity of $\hat{\varphi}(p, \cdot)$, we obtain $\hat{\varphi}(p, q) = \lim_{n, \mathcal{U}} \hat{\varphi}(p, \text{rtp}_\varphi(b_n)) = \lim_{n, \mathcal{U}} p(b_n) \leq r$.

So $s \leq \hat{\varphi}(p, q) \leq r$, contradicting $r < s$. $\odot$

The types p_m in the proof of Proposition 5.2.2 form a ladder in $\text{LS}(\varphi)$ against the b_n , which is exactly what the separate continuity of $\hat{\varphi}$ forbids: an independent sequence is shattered in every pattern, and a staircase is one of the patterns.

Stability was characterized in Section 1.2 by the exchange of iterated limits; NIP, which by Proposition 5.2.2 is the weaker condition, is characterized by the existence of convergent subsequences of the evaluation maps $e_b: \text{LS}(\varphi) \rightarrow [0, 1]$, given by $e_b(p) = p(b)$ for $b \in B$, as in Subsection 1.2.4. The analytic input is Rosenthal's dichotomy.

Fact 5.2.3 (Rosenthal). *Let X be a set and (f_n) a uniformly bounded sequence of real-valued functions on X . Then (f_n) has a subsequence that converges pointwise on X , or a subsequence (f_{n_k}) for which there are reals $r < s$ such that for all disjoint finite sets $P, Q \subseteq \omega$ there is $x \in X$ with $f_{n_k}(x) \geq s$ for $k \in P$ and $f_{n_k}(x) \leq r$ for $k \in Q$.*

Theorem 5.2.4. *For $\varphi: A \times B \rightarrow [0, 1]$ the following are equivalent.*

- (1) *The function φ is NIP.*
- (2) *For every sequence (b_n) in B , the sequence of functions (e_{b_n}) has a subsequence that converges pointwise on $\text{LS}(\varphi)$.*

Proof. (2) $\Rightarrow$ (1). Suppose that (b_n) is independent for φ , with reals $r < s$, and let (b_{n_k}) be a subsequence. For each $K < \omega$ choose $a_K \in A$ with $\varphi(a_K, b_{n_k}) \geq s$ for even $k < K$ and $\varphi(a_K, b_{n_k}) \leq r$ for odd $k < K$. Let $\mathcal{U}$ be a nonprincipal ultrafilter on ω and $p = \lim_{K, \mathcal{U}} \text{ltp}_\varphi(a_K)$, a point of $\text{LS}(\varphi)$. For each fixed k the inequality for k holds for all $K > k$, so $e_{b_{n_k}}(p) = p(b_{n_k})$ is at least s for even k and at most r for odd k . Hence $(e_{b_{n_k}}(p))_k$ does not converge. Since the subsequence was arbitrary, (2) fails.

(1) $\Rightarrow$ (2). Suppose (2) fails for (b_n) . By Fact 5.2.3 applied to $X = \text{LS}(\varphi)$ and $f_n = e_{b_n}$, there are a subsequence (b_{n_k}) and reals $r < s$ such that for all disjoint finite $P, Q \subseteq \omega$ there is $p \in \text{LS}(\varphi)$ with $p(b_{n_k}) \geq s$ for $k \in P$ and $p(b_{n_k}) \leq r$ for $k \in Q$. Fix $r' < s'$ with $r < r' < s' < s$. Since p is a pointwise limit of left types $\text{ltp}_\varphi(a)$ and only finitely many coordinates are involved, some $a \in A$ satisfies $\varphi(a, b_{n_k}) > s'$ for $k \in P$ and $\varphi(a, b_{n_k}) < r'$ for $k \in Q$. So (b_{n_k}) is independent for φ , with $r' < s'$, and φ is not NIP. $\odot$

The failure of condition (2) has a striking topological form: an independent sequence produces a copy of $\beta\mathbb{N}$.

Proposition 5.2.5. *If (b_n) is independent for φ , then the map $\mathcal{U} \mapsto \lim_{n, \mathcal{U}} e_{b_n}$ is a homeomorphism from $\beta\mathbb{N}$ onto a closed subset of the pointwise closure of $\{e_b : b \in B\}$ in $[0, 1]^{\text{LS}(\varphi)}$.*

Proof. The map $n \mapsto e_{b_n}$ from $\mathbb{N}$ into the compact space $[0, 1]^{\text{LS}(\varphi)}$ extends to the continuous map $\mathcal{U} \mapsto \lim_{n, \mathcal{U}} e_{b_n}$ on $\beta\mathbb{N}$, whose image lies in the pointwise closure of $\{e_b : b \in B\}$. Since $\beta\mathbb{N}$ is compact and $[0, 1]^{\text{LS}(\varphi)}$ is Hausdorff, it remains to be shown that the map is injective. Let $r < s$ witness the independence of (b_n) , let $\mathcal{U} \neq \mathcal{V}$ be ultrafilters on $\mathbb{N}$, and choose $X \in \mathcal{U}$ with $\mathbb{N} \setminus X \in \mathcal{V}$. For each finite $F \subseteq \mathbb{N}$ there is $a_F \in A$ with $\varphi(a_F, b_n) \geq s$ for $n \in F \cap X$ and $\varphi(a_F, b_n) \leq r$ for $n \in F \setminus X$. Let $\mathcal{D}$ be an ultrafilter on the finite subsets of $\mathbb{N}$ that contains $\{G : G \supseteq F\}$ for every finite F , and put $p = \lim_{F, \mathcal{D}} \text{ltp}_\varphi(a_F) \in \text{LS}(\varphi)$. Then $p(b_n) \geq s$ for $n \in X$ and $p(b_n) \leq r$ for $n \notin X$. Hence the limit of $e_{b_n}(p)$ along $\mathcal{U}$ is at least s , the limit along $\mathcal{V}$ is at most r , and the two limits of (e_{b_n}) differ at p . $\odot$

So the instances of φ , viewed as continuous functions on $\text{LS}(\varphi)$, face a dichotomy. If φ is NIP, every sequence of instances has a pointwise convergent subsequence (see Theorem 5.2.4); if φ has the independence property, the closure of the instances contains $\beta\mathbb{N}$, a compact space in which no sequence of distinct points converges. This is the topological face of Rosenthal's ℓ_1 dichotomy. In the theory of Bourgain, Fremlin and Talagrand [1], the same dichotomy separates the *Rosenthal compacta*, the compact spaces of Baire class 1 functions on Polish spaces, from compact spaces of functions that contain $\beta\mathbb{N}$, and Simon [14] develops the NIP side of this picture for spaces of types. Stable functions sit at the rigid end of the spectrum: their type spaces are Eberlein compacta (see Corollary A.1.8 below).

 **Remark 5.2.6.** Proposition 5.2.2 can also be derived, with heavier machinery, from Theorem 5.2.4. By Grothendieck's double limit criterion, in the form of Proposition A.1.7, stability of φ makes $\{e_b : b \in B\}$ relatively weakly compact in $C(\text{LS}(\varphi))$. By Corollary A.1.6, which rests on the Eberlein–Šmulian theorem (see Fact A.1.5 below), every sequence (e_{b_n}) then has a pointwise convergent subsequence. Thus condition (2) of Theorem 5.2.4 holds, and φ is NIP by the implication (2) $\Rightarrow$ (1), which does not use Rosenthal's dichotomy. The direct proof of Proposition 5.2.2 avoids both Grothendieck's criterion and the Eberlein–Šmulian theorem.

 **Remark 5.2.7.** Theorem 1.2.9 represents stable functions by the pairing of a reflexive Banach space E with E^* . Khanaki [5] proved that φ is stable, NIP, or NSOP (that is, without the strict order property of Remark 5.1.6) according as a Banach space naturally associated with φ is reflexive, contains no copy of ℓ_1 , or is weakly sequentially complete (every weakly Cauchy sequence converges weakly).

5.3. Historical remarks

 **Section 5.1.** The independence property was introduced by Shelah [36], in the same paper as the order property. His theorem that an unstable theory has the independence property or the strict order property is in [57, Chapter II]. The combinatorial notion of shattering, and the VC dimension, were introduced independently in probability and learning theory by Vapnik and Chervonenkis [15]; the polynomial bound on the number of traces (the subsets cut out on an n -element set) of a family of finite VC dimension was found independently by Sauer [11] and by Shelah [12]. Laskowski [6] observed that a formula is NIP exactly when the family of sets defined by the formula has finite VC dimension, connecting model theory with learning theory.

 **Section 5.2.** Rosenthal's dichotomy comes from his characterization of Banach spaces containing ℓ_1 [9], and the systematic study of pointwise compact sets of Baire class 1 functions is due to Rosenthal [10] and to Bourgain, Fremlin and Talagrand [1]. Their work entered model theory much later. The development is summarized in the following timeline.



1971 • **Shelah; Vapnik–Chervonenkis.** Shelah [36] introduces the independence property. Independently, Vapnik and Chervonenkis [15] introduce

shattering and what is now called VC dimension, in the study of uniform convergence of empirical frequencies.

- 1972

• **Sauer; Shelah.** Sauer [11] and Shelah [12] independently prove the polynomial bound on the number of traces of a set system of finite VC dimension.
- 1974–1978

• **Rosenthal; Bourgain–Fremlin–Talagrand.** Rosenthal proves the ℓ_1 dichotomy [9] and studies pointwise compact sets of Baire class 1 functions [10]. Bourgain, Fremlin and Talagrand [1] develop the theory of these “Rosenthal compacta”.
- 1981

• **Poizat.** Poizat [8] characterizes countable NIP theories by a bound on the number of types, and develops the theory of unstable theories without the independence property.
- 1986

• **Pillay–Steinhorn.** Pillay and Steinhorn [7] develop the theory of o-minimal structures, following van den Dries. These structures are NIP and are the main source of unstable NIP theories.
- 1992

• **Laskowski.** Laskowski [6] identifies NIP formulas with formulas defining families of finite VC dimension.
- 2008–2011

• **Hrushovski–Peterzil–Pillay; Hrushovski–Pillay.** The study of invariant and generically stable measures in NIP theories [3, 4], motivated by Pillay’s conjecture on definable groups in o-minimal structures, turns NIP into a central topic of model theory.
- 2009

• **Ben Yaacov.** Ben Yaacov [2] extends VC classes to $[0, 1]$ -valued functions, the continuous analogue of NIP used in Section 5.2.
- 2015

• **Simon.** Simon [14] applies the theory of Rosenthal compacta to spaces of φ -types of NIP formulas, and publishes the monograph [13] on NIP theories.
- 2015–2020

• **Khanaki.** Khanaki [20, 5] relates stability, NIP and NSOP to three properties of Banach spaces (see Remark 5.2.7).

Appendix A

Grothendieck's double limit criterion

Grothendieck's double limit criterion is used at several points in these notes: in Subsection 1.2.4 and in Remarks 1.2.14 and 5.2.6. This appendix states the criterion in the form we need and explains how the criterion relates to the stability of functions (see Definition 1.2.5). Throughout, we fix a compact Hausdorff space K , and $C(K)$ denotes the Banach space of continuous real-valued functions on K with the supremum norm. The *weak topology* on $C(K)$ is the coarsest topology making every bounded linear functional continuous, and the *pointwise topology* is the one induced by the product topology of $\mathbb{R}^K$. Point evaluations $f \mapsto f(x)$ are bounded linear functionals, so the weak topology is finer than the pointwise topology.

A.1. The criterion

Definition A.1.1. A bounded set $H \subseteq C(K)$ satisfies the *double limit condition* if for all sequences (x_m) in K and (f_n) in H and all ultrafilters $\mathcal{U}, \mathcal{V}$ on $\mathbb{N}$, we have

$$\lim_{m, \mathcal{U}} \lim_{n, \mathcal{V}} f_n(x_m) = \lim_{n, \mathcal{V}} \lim_{m, \mathcal{U}} f_n(x_m).$$

In the language of Definition 1.2.5, the double limit condition says that the evaluation function $K \times H \rightarrow \mathbb{R}$, $(x, f) \mapsto f(x)$, is stable. Grothendieck stated the condition with ordinary limits; the next lemma shows that the two formulations are equivalent.

Lemma A.1.2. *A bounded set $H \subseteq C(K)$ satisfies the double limit condition if and only if*

$$\lim_m \lim_n f_n(x_m) = \lim_n \lim_m f_n(x_m)$$

for all sequences (x_m) in K and (f_n) in H for which both iterated limits exist.

Proof. If both ordinary iterated limits exist, these limits are equal to the iterated limits along any nonprincipal ultrafilters, so the double limit condition implies the second condition.

Conversely, suppose the double limit condition fails for (x_m) , (f_n) , $\mathcal{U}$ and $\mathcal{V}$, with iterated limits $\alpha \neq \beta$. After an affine change of scale we may assume that all values lie in $[0, 1]$. Proposition 1.2.3, applied to the function $(x, f) \mapsto f(x)$, gives subsequences, which we still denote (x_m) and (f_n) , with $|f_n(x_m) - \alpha| \leq 2^{1-m}$ for $m < n$ and $|f_n(x_m) - \beta| \leq 2^{1-n}$ for $n < m$. By a diagonal argument we may pass to further subsequences along which $\lim_n f_n(x_m)$ exists for every m and $\lim_m f_n(x_m)$ exists for every n ; the inequalities are preserved, with the indices renumbered. Then $\lim_n f_n(x_m)$ is within 2^{1-m} of α , so $\lim_m \lim_n f_n(x_m) = \alpha$, and symmetrically $\lim_n \lim_m f_n(x_m) = \beta$. So the second condition fails. ☹

Theorem A.1.3 (Grothendieck). *Let $H \subseteq C(K)$ be bounded. The following are equivalent.*

- (1) *The set H is relatively weakly compact in $C(K)$.*
- (2) *Every function in the pointwise closure of H in $\mathbb{R}^K$ is continuous.*
- (3) *The set H satisfies the double limit condition.*

Proof. (2) $\Rightarrow$ (3). Let (x_m) , (f_n) , $\mathcal{U}$, $\mathcal{V}$ be given, and put $x = \lim_{m,\mathcal{U}} x_m \in K$ and $f = \lim_{n,\mathcal{V}} f_n \in \mathbb{R}^K$. By (2), the function f is continuous. Using the continuity of f and of each f_n , we obtain

$$\lim_{m,\mathcal{U}} \lim_{n,\mathcal{V}} f_n(x_m) = \lim_{m,\mathcal{U}} f(x_m) = f(x) = \lim_{n,\mathcal{V}} f_n(x) = \lim_{n,\mathcal{V}} \lim_{m,\mathcal{U}} f_n(x_m).$$

(3) $\Rightarrow$ (2). Suppose that f is in the pointwise closure of H but is not continuous at some $x \in K$. Then $f = \lim_{i,\mathcal{U}} f_i$ for a family $(f_i)_{i \in I}$ in H and an ultrafilter $\mathcal{U}$ on I , and there are a family $(x_j)_{j \in J}$ in K and an ultrafilter $\mathcal{V}$ on J with $\lim_{j,\mathcal{V}} x_j = x$ and $\lim_{j,\mathcal{V}} f(x_j) = \beta \neq f(x)$. (Fix $\varepsilon > 0$ such that every neighborhood O of x contains a point x_O with $|f(x_O) - f(x)| \geq \varepsilon$, and let $\mathcal{V}$ be an ultrafilter on the set of neighborhoods of x containing, for each O , the set of neighborhoods contained in O .) By continuity of each f_i , we have

$$\lim_{i,\mathcal{U}} \lim_{j,\mathcal{V}} f_i(x_j) = \lim_{i,\mathcal{U}} f_i(x) = f(x), \quad \lim_{j,\mathcal{V}} \lim_{i,\mathcal{U}} f_i(x_j) = \lim_{j,\mathcal{V}} f(x_j) = \beta.$$

After an affine change of scale, Proposition 1.2.3 and the proof of Corollary 1.2.6 turn these families into sequences whose iterated limits along a nonprincipal ultrafilter are β and $f(x)$. This contradicts (3).

(1) $\Rightarrow$ (2). Let W be the weak closure of H , which is weakly compact by (1). The identity map from $C(K)$ with the weak topology to $C(K)$ with the pointwise topology is continuous, so W is pointwise compact, hence closed in $\mathbb{R}^K$. So the pointwise closure of H is contained in $W \subseteq C(K)$.

(2) $\Rightarrow$ (1). Let $\bar{H}$ be the pointwise closure of H . This set is compact in $\mathbb{R}^K$, because H is bounded, and is contained in $C(K)$ by (2). We need only show that the weak and pointwise topologies agree on $\bar{H}$, since then $\bar{H}$ is weakly compact. By the Riesz representation theorem, the weak topology is generated by the maps $h \mapsto \int h d\mu$ with μ a Radon measure on K . On the uniformly bounded set $\bar{H}$ each such map is sequentially continuous for the pointwise topology, by Lebesgue's dominated convergence theorem. Grothendieck proved that a pointwise compact subset of $C(K)$ is *angelic*: every point in the closure of a subset is the limit of a sequence from that subset [44]; see also Pryce [25]. On an angelic space sequentially continuous functions are continuous, which completes the proof. ☺

Example A.1.4. Let $K = [0, 1]$ and $H = \{f_n : n \geq 1\}$ with $f_n(x) = x^n$. The pointwise limit of (f_n) is 0 on $[0, 1)$ and 1 at 1, which is not continuous, so (2) of Theorem A.1.3 fails. Condition (3) fails along $x_m = 1 - 1/m$: for each m we have $\lim_n f_n(x_m) = 0$, while for each n we have $\lim_m f_n(x_m) = 1$, so the two iterated limits are 0 and 1 (see Figure A.1). Hence H is not relatively weakly compact in $C[0, 1]$.

The proof of (2) $\Leftrightarrow$ (3) in Theorem A.1.3 uses only ultrafilter limits and Proposition 1.2.3. The measure-theoretic input and the angelicity of pointwise compact sets enter only in (2) $\Rightarrow$ (1).

Fact A.1.5 (Eberlein–Šmulian). *In a Banach space, every sequence in a relatively weakly compact set has a weakly convergent subsequence [16, 17].*

Corollary A.1.6. *If a bounded set $H \subseteq C(K)$ satisfies the double limit condition, then every sequence in H has a subsequence that converges pointwise on K , to a continuous function.*

Proof. By Theorem A.1.3, the set H is relatively weakly compact. By Fact A.1.5, every sequence in H has a subsequence converging weakly to some $f \in C(K)$. Point evaluations are bounded linear functionals, so the subsequence converges pointwise to f . ☺

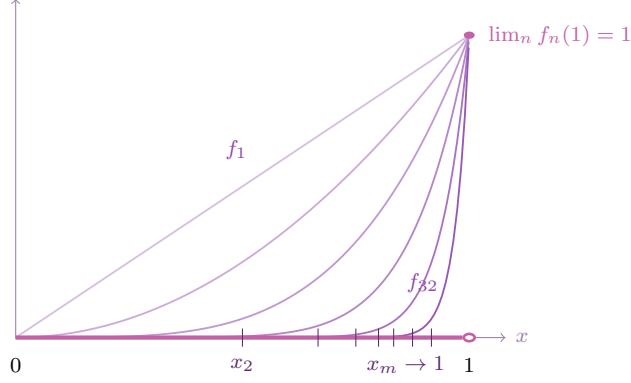


FIGURE A.1. Example A.1.4: the functions $f_n(x) = x^n$ (purple, darker for larger n) converge pointwise to a function (pink) that is 0 on $[0, 1)$ and jumps to 1 at $x = 1$. Along the points $x_m = 1 - 1/m$, letting $n \rightarrow \infty$ first gives 0, and letting $m \rightarrow \infty$ first gives 1.

The criterion meets stability through the spaces of types of Section 1.2. For $\varphi: A \times B \rightarrow [0, 1]$, recall the evaluation maps $e_b: \text{LS}(\varphi) \rightarrow [0, 1]$, given by $e_b(p) = p(b)$, and $e_a: \text{RS}(\varphi) \rightarrow [0, 1]$, given by $e_a(q) = q(a)$, of Subsection 1.2.4.

Proposition A.1.7. *Let $\varphi: A \times B \rightarrow [0, 1]$ be stable. Then $\{e_b : b \in B\}$ is relatively weakly compact in $C(\text{LS}(\varphi))$, and $\{e_a : a \in A\}$ is relatively weakly compact in $C(\text{RS}(\varphi))$. More precisely, the pointwise closure of $\{e_b : b \in B\}$ consists of the functions $\hat{\varphi}(\cdot, q)$ with $q \in \text{RS}(\varphi)$, where $\hat{\varphi}$ is the function of Proposition 1.2.8.*

Proof. By (1.2) with a constant family, we have $e_b = \hat{\varphi}(\cdot, \text{rtp}_\varphi(b))$. If $q = \lim_{j, \gamma} \text{rtp}_\varphi(b_j)$, then by continuity of $\hat{\varphi}(p, \cdot)$ we get $\lim_{j, \gamma} e_{b_j}(p) = \hat{\varphi}(p, q)$ for every p . So every pointwise limit of functions e_b is of the form $\hat{\varphi}(\cdot, q)$, and conversely every such function is a pointwise limit of this kind. These functions are continuous on $\text{LS}(\varphi)$ by separate continuity. Hence condition (2) of Theorem A.1.3 holds, and $\{e_b : b \in B\}$ is relatively weakly compact. The statement for $\{e_a : a \in A\}$ is proved symmetrically. ☺

Proposition A.1.7 places the type spaces of a stable function among the most rigid compact spaces of topology. Recall that a compact space is an *Eberlein compactum* if the space is homeomorphic to a weakly compact subset of a Banach space with the weak topology.

Corollary A.1.8. *Let $\varphi: A \times B \rightarrow [0, 1]$ be stable. Then $\text{RS}(\varphi)$ and $\text{LS}(\varphi)$ are Eberlein compacta. In particular, every right φ -type is the limit of a sequence of realized right types, and likewise for left types.*

Proof. Let H be the pointwise closure of $\{e_b : b \in B\}$ in $\mathbb{R}^{\text{LS}(\varphi)}$. By Proposition A.1.7, the weak closure W of $\{e_b : b \in B\}$ in $C(\text{LS}(\varphi))$ is weakly compact. The identity map from W with the weak topology to W with the coarser pointwise topology is a continuous bijection from a compact space to a Hausdorff space, so the two topologies agree on W . In particular, the set W is pointwise compact, hence pointwise closed, so $H \subseteq W$; and $W \subseteq H$ because the weak topology is finer than the pointwise topology. Thus $H = W$, and on H the weak and pointwise topologies agree. The map $q \mapsto \hat{\varphi}(\cdot, q)$ from $\text{RS}(\varphi)$ onto H is continuous for the pointwise topology, by the separate continuity of $\hat{\varphi}$, and injective, because $\hat{\varphi}(\text{ltp}_\varphi(a), q) = q(a)$ for every $a \in A$. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism, so $\text{RS}(\varphi)$ is homeomorphic to the weakly compact set

H. The same argument, with the roles of A and B exchanged, applies to $\text{LS}(\varphi)$. Finally, the set H is a pointwise compact subset of $C(\text{LS}(\varphi))$, hence angelic (see the discussion before Fact A.1.5), so each point of H is the limit of a sequence from $\{e_b : b \in B\}$; through the homeomorphism, each $q \in \text{RS}(\varphi)$ is the limit of a sequence of realized types $\text{rtp}_\varphi(b_n)$. ☺

Sequences are a genuine restriction. In the unstable structure $(\omega_1, <)$ with $\varphi(x, y)$ the formula $x < y$, the left type p given by $p(\beta) = 0$ for every $\beta < \omega_1$ (an element above every countable ordinal) is a limit of realized types along an ultrafilter, but not along any sequence: if $\alpha_n < \omega_1$ for every n , then $\alpha = \sup_n \alpha_n < \omega_1$, and the realized types $\text{ltp}_\varphi(\alpha_n)$ all take the value 1 at $\alpha + 1$, so no subsequence of them converges to p .

A.2. Historical remarks

⌘ **Section A.1.** Theorem A.1.3 is from Grothendieck's 1952 paper [44], announced in [18], which also proves that pointwise compact subsets of $C(K)$ are angelic. The Eberlein–Šmulian theorem is due to Šmulian [16] and Eberlein [17]. Later proofs and extensions of the criterion, by Namioka, Pták, Young, Simons, Pryce and König, are listed in the timeline in the historical remarks on Subsection 1.2.1. The reading of the criterion as a statement about stable functions, which is the point of view of Definition A.1.1 and Proposition A.1.7, is due to Ben Yaacov [42] and Pillay [21] for formulas, following Iovino [49] and Krivine–Maurey [51].

Bibliography

- [1] J. Bourgain, D. H. Fremlin and M. Talagrand, *Pointwise compact sets of Baire-measurable functions*, Amer. J. Math. **100** (1978), 845–886.
- [2] I. Ben Yaacov, *Continuous and random Vapnik–Chervonenkis classes*, Israel J. Math. **173** (2009), 309–333.
- [3] E. Hrushovski, Y. Peterzil and A. Pillay, *Groups, measures, and the NIP*, J. Amer. Math. Soc. **21** (2008), 563–596.
- [4] E. Hrushovski and A. Pillay, *On NIP and invariant measures*, J. Eur. Math. Soc. **13** (2011), 1005–1061.
- [5] K. Khanaki, *Correspondences between model theory and Banach space theory*, arXiv:1512.08691.
- [6] M. C. Laskowski, *Vapnik–Chervonenkis classes of definable sets*, J. London Math. Soc. (2) **45** (1992), 377–384.
- [7] A. Pillay and C. Steinhorn, *Definable sets in ordered structures. I*, Trans. Amer. Math. Soc. **295** (1986), 565–592.
- [8] B. Poizat, *Théories instables*, J. Symbolic Logic **46** (1981), 513–522.
- [9] H. P. Rosenthal, *A characterization of Banach spaces containing ℓ^1* , Proc. Nat. Acad. Sci. U.S.A. **71** (1974), 2411–2413.
- [10] H. P. Rosenthal, *Point-wise compact subsets of the first Baire class*, Amer. J. Math. **99** (1977), 362–378.
- [11] N. Sauer, *On the density of families of sets*, J. Combin. Theory Ser. A **13** (1972), 145–147.
- [12] S. Shelah, *A combinatorial problem; stability and order for models and theories in infinitary languages*, Pacific J. Math. **41** (1972), 247–261.
- [13] P. Simon, *A Guide to NIP Theories*, Lecture Notes in Logic 44, Cambridge University Press, 2015.
- [14] P. Simon, *Rosenthal compacta and NIP formulas*, Fund. Math. **231** (2015), 81–92.
- [15] V. N. Vapnik and A. Ya. Chervonenkis, *On the uniform convergence of relative frequencies of events to their probabilities*, Theory Probab. Appl. **16** (1971), 264–280.
- [16] V. Šmulian, *Über lineare topologische Räume*, Mat. Sbornik N.S. **7(49)** (1940), 425–448.
- [17] W. F. Eberlein, *Weak compactness in Banach spaces. I*, Proc. Nat. Acad. Sci. U.S.A. **33** (1947), 51–53.
- [18] A. Grothendieck, note in C. R. Acad. Sci. Paris **230** (1950), 605–606.
- [19] E. Hrushovski and A. Pillay, *Groups definable in local fields and pseudo-finite fields*, Israel J. Math. **85** (1994), 203–262.
- [20] K. Khanaki, *Stability, the NIP, and the NSOP: model theoretic properties of formulas via topological properties of function spaces*, Math. Logic Quarterly (2020), doi:10.1002/malq.201500059; arXiv:1410.3339.
- [21] A. Pillay, *Stability and Grothendieck*, arXiv:1703.04761 (2017).
- [22] H. König, *Angelic spaces and the double limit relation*, J. London Math. Soc. (2) **35** (1987), 454–470.
- [23] I. Namioka, *A substitute for Lebesgue’s bounded convergence theorem*, Proc. Amer. Math. Soc. **12** (1961), 713–716.
- [24] I. Namioka, *Separate continuity and joint continuity*, Pacific J. Math. **51** (1974), 515–531.
- [25] J. D. Pryce, *A device of R. J. Whitley’s applied to pointwise compactness in spaces of continuous functions*, Proc. London Math. Soc. (3) **23** (1971), 532–546.
- [26] V. Pták, *A combinatorial lemma on the existence of convex means and its application to weak compactness*, in: Convexity, Proc. Sympos. Pure Math. **7**, Amer. Math. Soc., 1963, 437–450.
- [27] V. Pták, *An extension theorem for separately continuous functions and its application to functional analysis*, Czechoslovak Math. J. **14(89)** (1964), 562–581.
- [28] S. Simons, *The iterated limit condition, a Fubini theorem and weak compactness*, Math. Ann. **176** (1968), 87–95.
- [29] S. Simons, *A convergence theorem with boundary*, Pacific J. Math. **40** (1972), 703–708.
- [30] S. Simons, *Maximinimax, minimax, and antiminimax theorems and a result of R. C. James*, Pacific J. Math. **40** (1972), 709–718.

- [31] S. Simons, *On Pták's combinatorial lemma*, Pacific J. Math. **40** (1972), 719–721.
- [32] N. J. Young, *On Pták's double-limit theorems*, Proc. Edinburgh Math. Soc. (2) **17** (1971), 193–200.
- [33] D. Lascar and B. Poizat, *An introduction to forking*, J. Symbolic Logic **44** (1979), 330–350.
- [34] S. Mazur, *Über konvexe Mengen in linearen normierten Räumen*, Studia Math. **4** (1933), 70–84.
- [35] M. Morley, *Categoricity in power*, Trans. Amer. Math. Soc. **114** (1965), 514–538.
- [36] S. Shelah, *Stability, the f.c.p., and superstability; model theoretic properties of formulas in first order theory*, Ann. Math. Logic **3** (1971), 271–362.
- [37] M. H. Stone, *Applications of the theory of Boolean rings to general topology*, Trans. Amer. Math. Soc. **41** (1937), 375–481.
- [38] M. H. Stone, *The generalized Weierstrass approximation theorem*, Math. Mag. **21** (1948), 167–184 and 237–254.
- [39] K. Weierstrass, *Über die analytische Darstellbarkeit sogenannter willkürlicher Functionen einer reellen Veränderlichen*, Sitzungsber. Königl. Preuß. Akad. Wiss. Berlin (1885), 633–639 and 789–805.
- [40] H. Gaifman, *Models and types of Peano's arithmetic*, Ann. Math. Logic **9** (1976), 223–306.
- [41] S. Shelah, *Stable theories*, Israel J. Math. **7** (1969), 187–202.
- [42] I. Ben Yaacov, *Model theoretic stability and definability of types, after A. Grothendieck*, Bull. Symbolic Logic **20** (2014), no. 4, 491–496.
- [43] I. Ben Yaacov and A. Usvyatsov, *Continuous first order logic and local stability*, Trans. Amer. Math. Soc. **362** (2010), no. 10, 5213–5259.
- [44] A. Grothendieck, *Critères de compacité dans les espaces fonctionnels généraux*, Amer. J. Math. **74** (1952), 168–186.
- [45] G. Conant and C. Terry, *Encoding orders and trees in real-valued functions*, arXiv:2607.21761.
- [46] W. J. Davis, T. Figiel, W. B. Johnson and A. Pełczyński, *Factoring weakly compact operators*, J. Funct. Anal. **17** (1974), 311–327.
- [47] J. Iovino, *Stable theories in functional analysis*, Ph.D. thesis, University of Illinois at Urbana-Champaign, 1994.
- [48] M. Makkai, *A survey of basic stability theory, with particular emphasis on orthogonality and regular types*, Israel J. Math. **49** (1984), 181–238.
- [49] J. Iovino, *Stable models and reflexive Banach spaces*, J. Symbolic Logic **64** (1999), no. 4, 1595–1600.
- [50] R. C. James, *Uniformly non-square Banach spaces*, Ann. of Math. **80** (1964), 542–550.
- [51] J.-L. Krivine and B. Maurey, *Espaces de Banach stables*, Israel J. Math. **39** (1981), 273–295.
- [52] Y. Raynaud, *Espaces de Banach superstables, distances stables et homéomorphismes uniformes*, Israel J. Math. **44** (1983), 33–52.
- [53] W. Hodges, *Model Theory*, Encyclopedia of Mathematics and its Applications 42, Cambridge University Press, 1993.
- [54] V. Harnik and L. Harrington, *Fundamentals of forking*, Ann. Pure Appl. Logic **26** (1984), 245–286.
- [55] B. Kim and A. Pillay, *Simple theories*, Ann. Pure Appl. Logic **88** (1997), 149–164.
- [56] A. Pillay, *Geometric Stability Theory*, Oxford Logic Guides 32, Oxford University Press, 1996.
- [57] S. Shelah, *Classification Theory and the Number of Non-isomorphic Models*, 2nd ed., North-Holland, 1990.
- [58] K. Tent and M. Ziegler, *A Course in Model Theory*, Lecture Notes in Logic 40, Cambridge University Press, 2012.
- [59] E. Steinitz, *Algebraische Theorie der Körper*, J. Reine Angew. Math. **137** (1910), 167–309.
- [60] B. L. van der Waerden, *Moderne Algebra*, Vol. I, Springer, Berlin, 1930.
- [61] H. Whitney, *On the abstract properties of linear dependence*, Amer. J. Math. **57** (1935), 509–533.
- [62] G. Birkhoff, *Abstract linear dependence and lattices*, Amer. J. Math. **57** (1935), 800–804.
- [63] S. Mac Lane, *Some interpretations of abstract linear dependence in terms of projective geometry*, Amer. J. Math. **58** (1936), 236–240.
- [64] H. H. Crapo and G.-C. Rota, *On the Foundations of Combinatorial Theory: Combinatorial Geometries*, preliminary edition, MIT Press, Cambridge, MA, 1970.
- [65] J. T. Baldwin and A. H. Lachlan, *On strongly categorical theories*, J. Symbolic Logic **36** (1971), 79–96.
- [66] F. Rowbottom, *The Loś conjecture for uncountable theories*, Notices Amer. Math. Soc. **11** (1964), 248 (abstract).